

# UNIT-III

## **SUPERVISED LEARNING**

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## UNIT-III

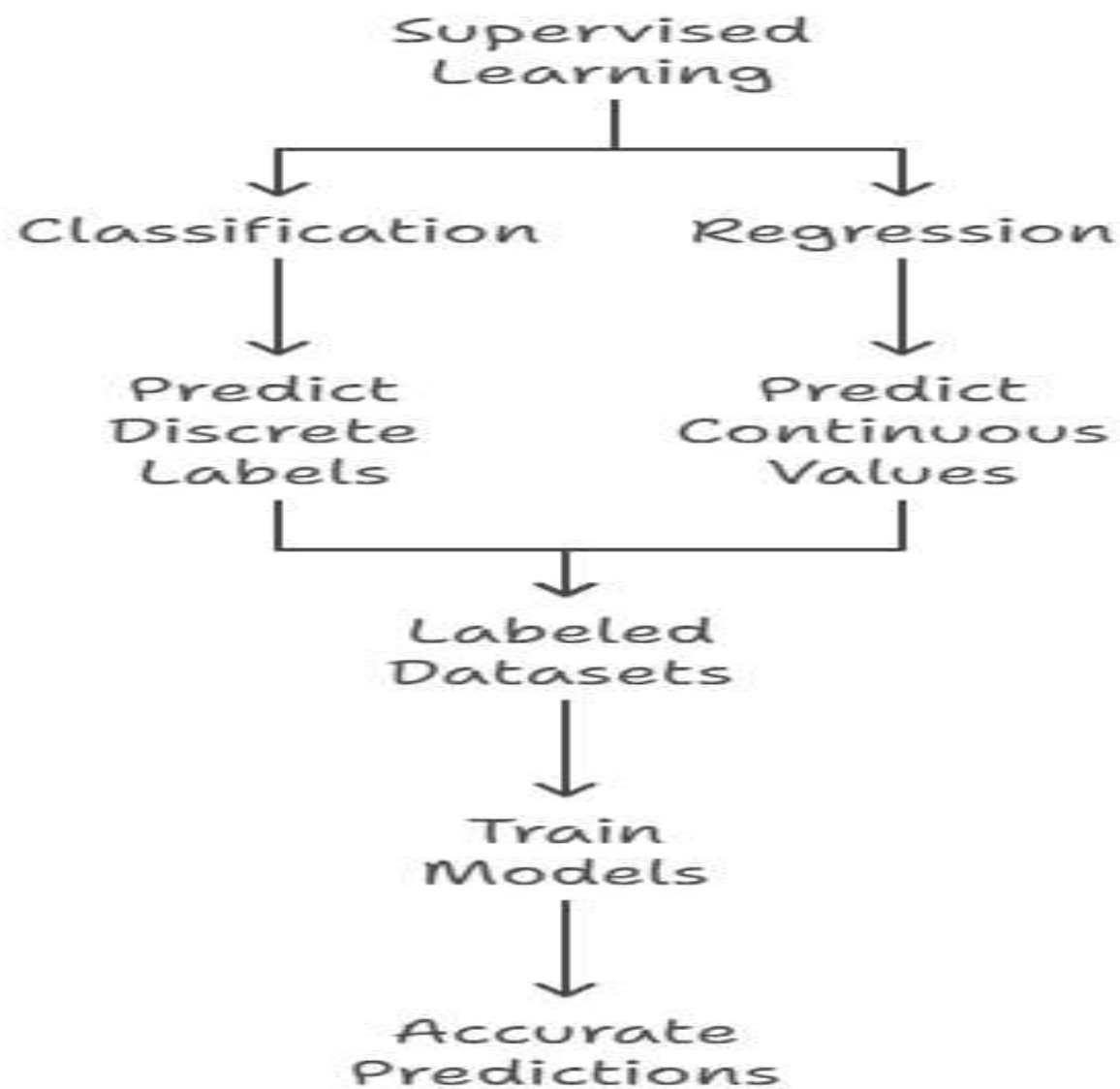
# SUPERVISED LEARNING

- Linear Regression -Relation between two variables – Steps – Evaluation – Logistic Regression – Decision Tree – Algorithms – Construction – Classification using Decision Tree – Issues – Rule - based Classification – Pruning the Rule Set – Support Vector Machines – Linear SVM – Optimal Hyperplane – Radial Basis Functions – Naïve Bayes Classifier – Bayesian Belief Networks.

### **Lab Programs:**

- Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select the appropriate data set for your experiment and draw graphs.
- Implement and demonstrate the working of the decision tree-based ID3 algorithm
- Build a Simple Support Vector Machines using a data set

# Supervised Learning Tasks



# Regression

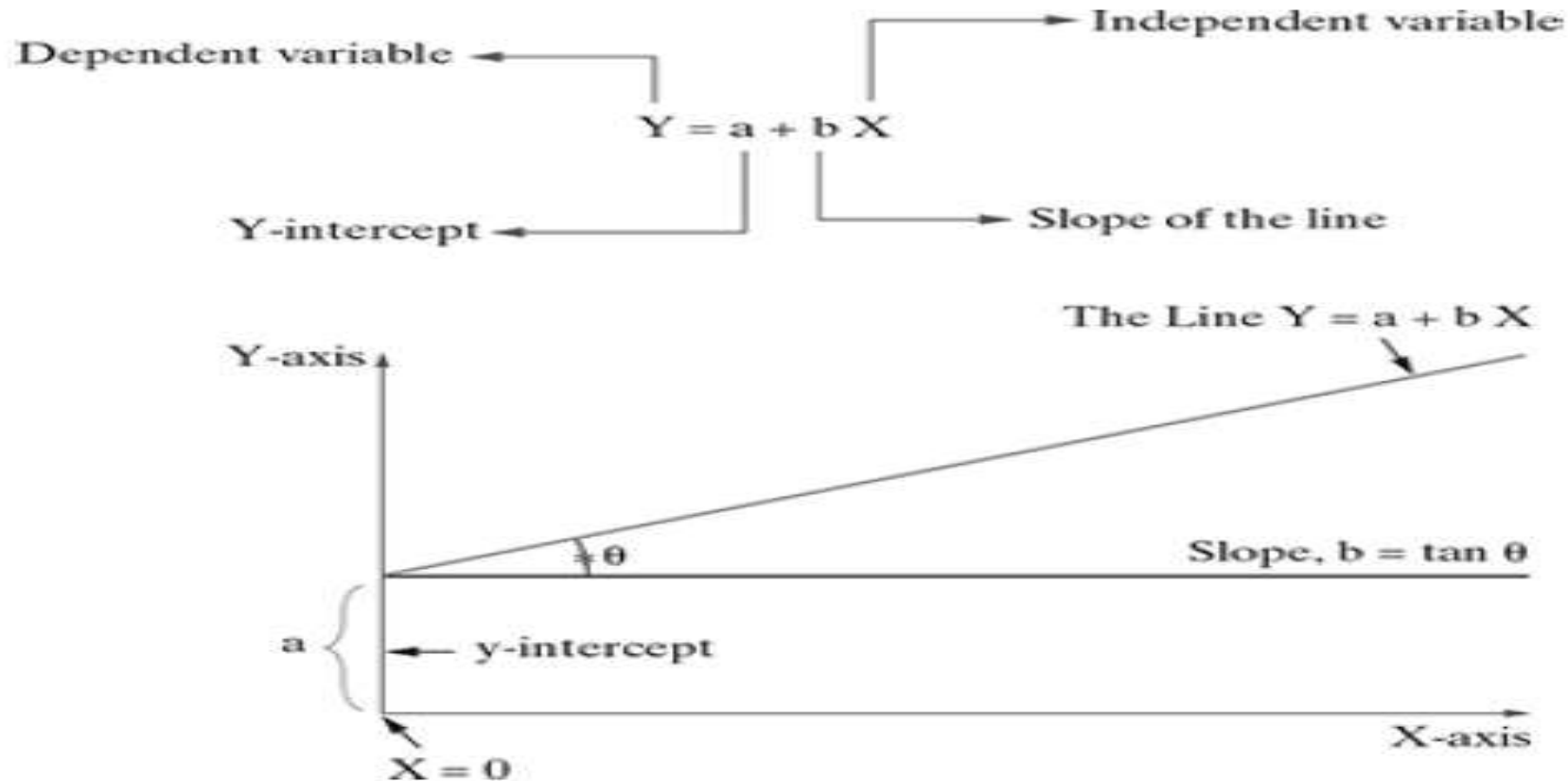
- Prediction of numerical variables
- Regression aims to **find** the "best fit" line or curve through the data points to forecast trends and specific values, such as predicting house prices, stock market performance, or patient health outcomes.
- Regression is essentially finding a relationship (or) association between the dependent variable ( $Y$ ) and the independent variable(s) ( $X$ ), i.e. to find the function ' $f$ ' for the association  $Y = f(X)$ .

# COMMON REGRESSION ALGORITHMS

- Simple linear regression
- Multiple linear regression
- Polynomial regression
- Multivariate adaptive regression splines
- Maximum likelihood estimation (least squares)

# Simple linear regression

- The simplest regression model which involves only one predictor.
- This model assumes a linear relationship between the dependent variable and the predictor variable.  $Y = a + bX$



- If we know the values of ' $a$ ' and ' $b$ ', then it is easy to predict the value of  $Y$  for any given  $X$  by using the above formula.
- How to calculate the values of ' $a$ ' and ' $b$ ' for a given set of  $X$  and  $Y$  values?
- **Ordinary Least Squares (OLS)** is the technique used to estimate a line that will minimize the error ( $\epsilon$ ), which is the difference between the predicted and the actual values of  $Y$ .

$$b = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{\text{cov}(X, Y)}{\text{Var}(X)}$$

$$a = \bar{Y} - b\bar{X}$$

# *OLS algorithm*

- Step 1: Calculate the mean of  $X$  and  $Y$
- Step 2: Calculate the errors of  $X$  and  $Y$
- Step 3: Get the product
- Step 4: Get the summation of the products
- Step 5: Square the difference of  $X$
- Step 6: Get the sum of the squared difference
- Step 7: Divide output of step 4 by output of step 6 to calculate ' $b$ '
- Step 8: Calculate ' $a$ ' using the value of ' $b$ '



# EXAMPLE

- A college professor believes that if the grade for internal examination is high in a class, the grade for external examination will also be high.
- A random sample of 15 students in that class was selected, and the data is given below: Calculate the slope and intercept using least squares.

|               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Internal Exam | 15 | 23 | 18 | 23 | 24 | 22 | 22 | 19 | 19 | 16 | 24 | 11 | 24 | 16 | 23 |
| External Exam | 49 | 63 | 58 | 60 | 58 | 61 | 60 | 63 | 60 | 52 | 62 | 30 | 59 | 49 | 68 |

# Calculation of regression parameters

|               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Internal Exam | 15 | 23 | 18 | 23 | 24 | 22 | 22 | 19 | 19 | 16 | 24 | 11 | 24 | 16 | 23 |
| External Exam | 49 | 63 | 58 | 60 | 58 | 61 | 60 | 63 | 60 | 52 | 62 | 30 | 59 | 49 | 68 |

Step 2

Step 3

Step 5

| X           | Y           | X- mean (X) | Y- Mean (Y) | $(X_i - \bar{X})(Y_i - \bar{Y})$                         | $(X_i - \bar{X})^2$ |
|-------------|-------------|-------------|-------------|----------------------------------------------------------|---------------------|
| 15          | 49          | -4.93       | -7.8        | 38.454                                                   | 24.3049             |
| 23          | 63          | 3.07        | 6.2         | 19.034                                                   | 9.4249              |
| 18          | 58          | -1.93       | 1.2         | -2.316                                                   | 3.7249              |
| 23          | 60          | 3.07        | 3.2         | 9.824                                                    | 9.4249              |
| 24          | 58          | 4.07        | 1.2         | 4.884                                                    | 16.5649             |
| 22          | 61          | 2.07        | 4.2         | 8.694                                                    | 4.2849              |
| 22          | 60          | 2.07        | 3.2         | 6.624                                                    | 4.2849              |
| 19          | 63          | -0.93       | 6.2         | -5.766                                                   | 0.8649              |
| 19          | 60          | -0.93       | 3.2         | -2.976                                                   | 0.8649              |
| 16          | 52          | -3.93       | -4.8        | 18.864                                                   | 15.4449             |
| 24          | 62          | 4.07        | 5.2         | 21.164                                                   | 16.5649             |
| 11          | 30          | -8.93       | -26.8       | 239.324                                                  | 79.7449             |
| 24          | 59          | 4.07        | 2.2         | 8.954                                                    | 16.5649             |
| 16          | 49          | -3.93       | -7.8        | 30.654                                                   | 15.4449             |
| 23          | 68          | 3.07        | 11.2        | 34.384                                                   | 9.4249              |
| <b>19.9</b> | <b>56.8</b> |             |             | <b><math>\Sigma(X_i - \bar{X})(Y_i - \bar{Y})</math></b> | <b>226.9335</b>     |

Step 4

Step 6

Step 8: Calculate a using the value of b

$$a = \bar{Y} - b\bar{X}$$

$$a = 56.8 - 1.89 \times 19.9$$

$$a = 19.05$$

*The predicted value can be calculated from the equation:*

$$\hat{y} = 1.89395X + 19.0473$$

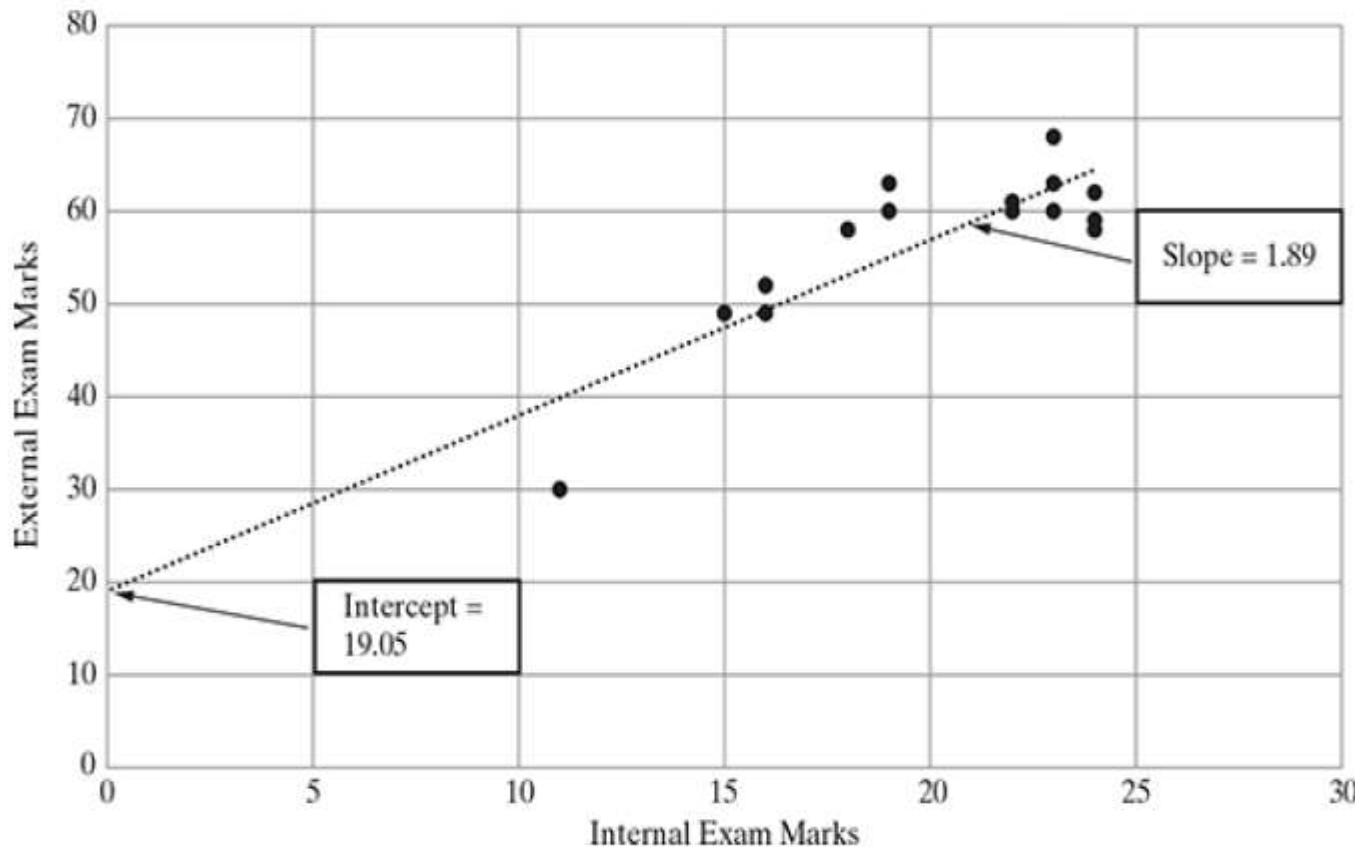
Step 1

Step 7: Divide (step4 / step6)

$$b = 429.28 / 226.93 = 1.89$$

So, in the context of the given problem, we can say

Marks in external exam =  $19.04 + 1.89 \times (\text{Marks in internal exam})$



Slope measures the estimated change in the average value of Y as a result of a one-unit change in X.

Intercept = 19.05 indicates that 19.05 is the portion of the external examination marks not explained by the internal examination marks.

# Multiple Linear Regression

- two or more independent variables, i.e. predictors are involved in the model
- if we consider Price of a Property (in \$) as the dependent variable and Area of the Property (in sq. m.), location, floor, number of years since purchase and amenities available as the independent variables

$$\hat{Y} = a + b_1X_1 + b_2X_2 + b_3X_3 + \dots + b_nX_n$$

# Polynomial regression model

- It is the extension of the simple linear model by adding extra predictors obtained by raising (squaring) each of the original predictors to a power.

$$f(x) = c_0 + c_1.X^1 + c_2.X^2 + c_3.X^3$$

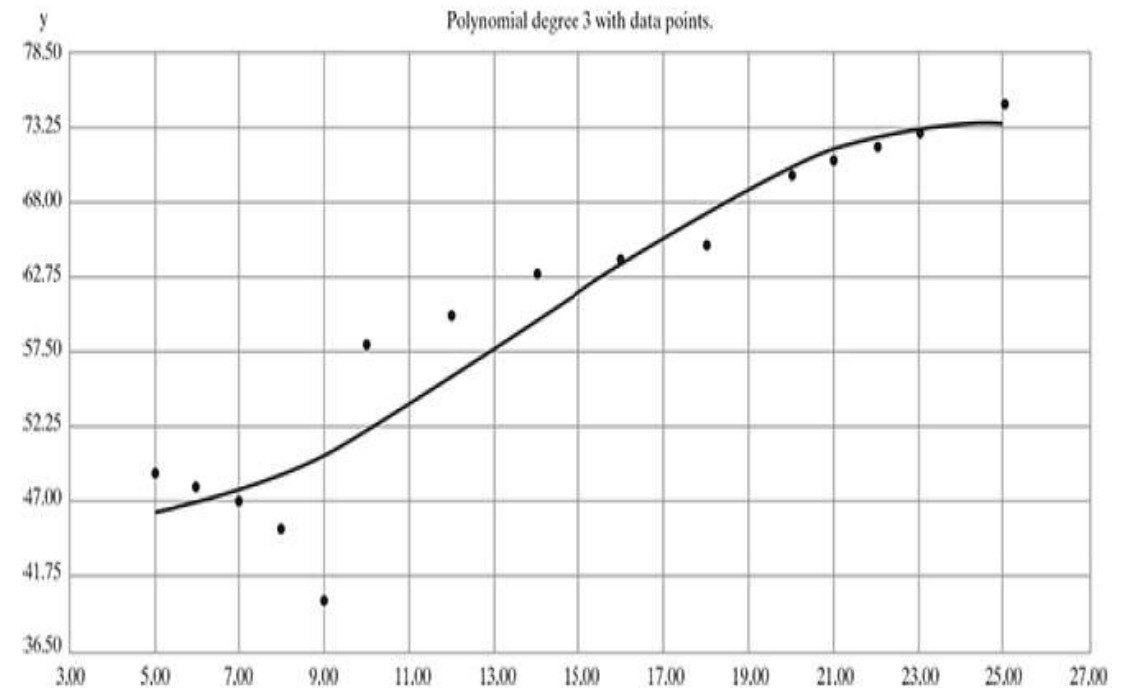


FIG. 8.16 Polynomial regression degree 3

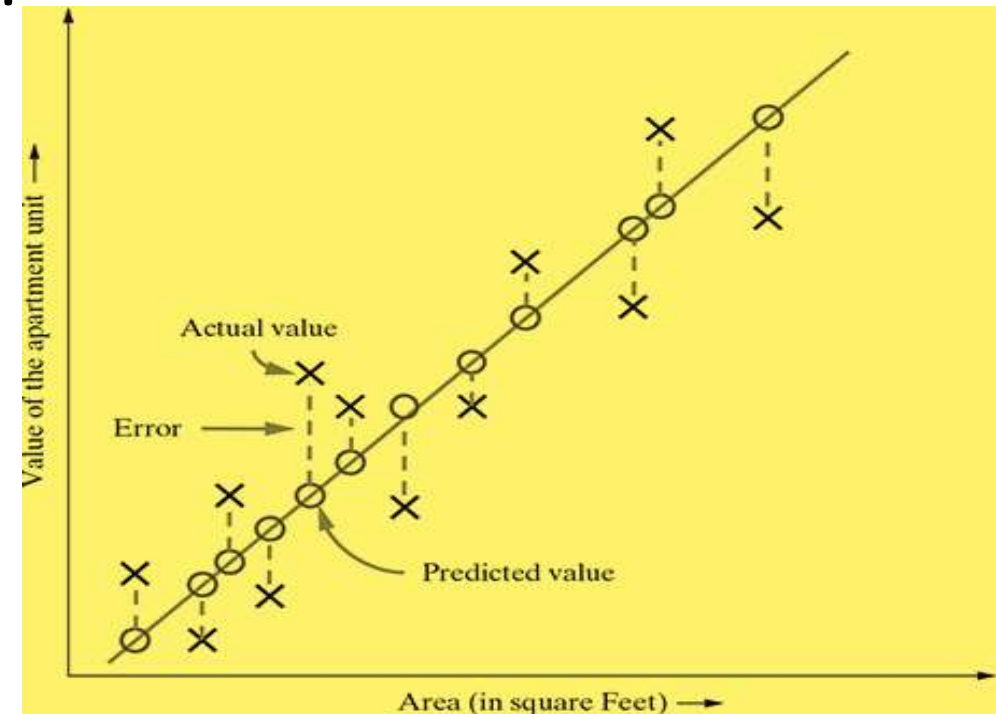
# EVALUATING PERFORMANCE OF Regression Model

- The performance of a regression model is evaluated by measuring how well it predicts continuous output variables.
- Several quantitative metrics are used to assess the quality of the predictions compared to the actual values.
- **Mean Absolute Error (MAE)**

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- **Mean Squared Error (MSE)**

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



- **Root Mean Squared Error (RMSE)**

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **R-squared ( $R^2$ )**

- Proportion of the variance in the dependent variable that is predictable from the independent variables.
- Range: 0 to 1 (can be negative if model performs worse than mean prediction).
- Interpretation: Closer to 1 means better model

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

## **Adjusted R-squared**

- Modified version of  $R^2$  that adjusts for the number of predictors in the model.
- Useful when comparing models with different numbers of features.

# Logistic Regression

- Logistic Regression is used for classification problems
- Predicts categorical outcomes (Yes/No, 0/1, Spam/Not Spam)
- Extension of Linear Regression with sigmoid transformation



# Why Not Linear Regression for Classification?

- Linear Regression outputs continuous values
- Predictions can be  $<0$  or  $>1$  (invalid probabilities)
- Logistic Regression uses Sigmoid Function to fix this

# Sigmoid Function

- Formula:

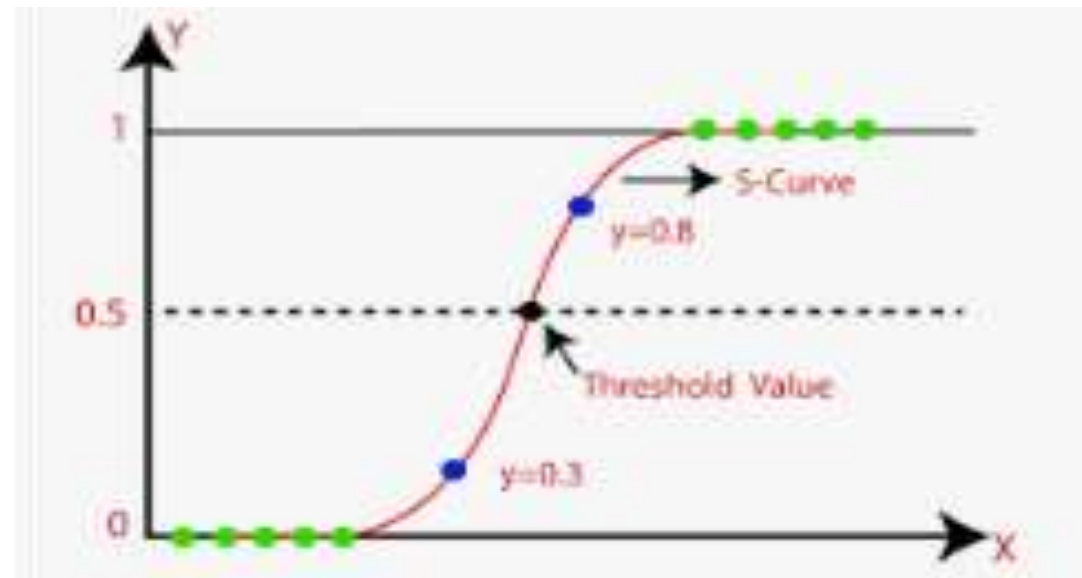
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

- Output always between 0 and 1
- Interpreted as probability of belonging to a class

# Logistic Regression

- Probability:  $P(y=1 | x) = \sigma(z)$
- Decision Rule: If  $P \geq 0.5 \rightarrow$  Class 1, else  $\rightarrow$  Class 0



# Cost Function (Log Loss)

- Linear Regression uses MSE → not suitable for classification
- Logistic Regression uses Log Loss (Cross Entropy Loss)

$$J(\theta) = -\frac{1}{m} \sum [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

# Types of Logistic Regression

- Binary Logistic Regression – 2 outcomes (Yes/No, 0/1)
- Multinomial Logistic Regression – multiple classes
- Ordinal Logistic Regression – ordered categories

# Applications

- Medical Diagnosis (disease vs. no disease)
- Email Spam Detection
- Customer Churn Prediction
- Credit Risk Assessment

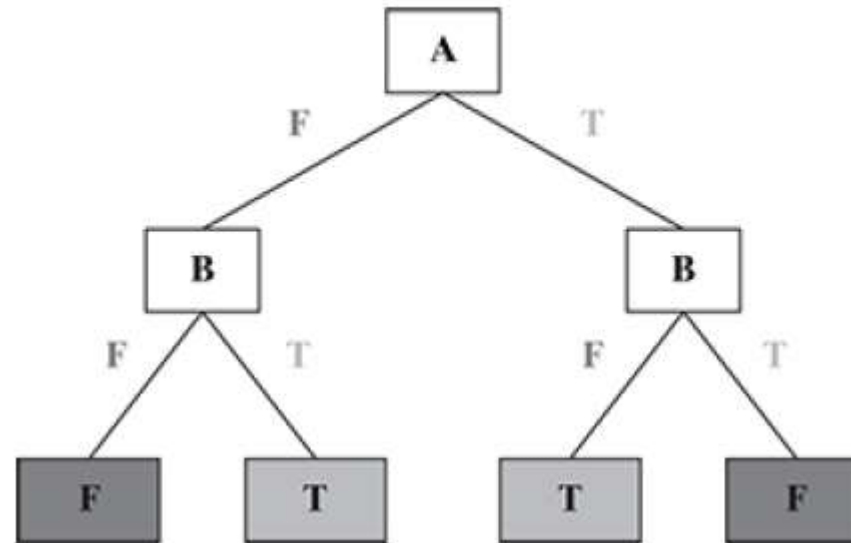
# Advantages & Limitations

- Advantages: Simple, interpretable, works with linearly separable data, probabilistic output
- Limitations: Assumes linear relationship in log-odds, not suitable for complex nonlinear data

# Decision Tree

A decision tree consists of three types of nodes:

- Root Node
- Branch Node
- Leaf Node





# Decision Tree

- There are many implementations of decision tree, the most prominent algorithms are:
  - ❖ ID3 (Iterative Dichotomiser 3)
  - ❖ C5.0
  - ❖ CART (Classification and Regression Tree),
  - ❖ CHAID (Chi-square Automatic Interaction Detector)
- The biggest challenge of a decision tree algorithm is to find out **which feature to split** upon.
- The main driver for identifying the feature is that the data should be split in such a way that the partitions created by the split should contain examples belonging to a single class. If that happens, the partitions are considered to be **pure**.

# Decision Tree Algorithm - Phases

- Decision tree generation consists of **two phases**

## Phase 1 : Tree construction

- At start, all the training examples are at the root
- Attributes are categorical (if continuous-valued, they are discretized in advance)
- Test attributes are selected on the basis of a heuristic or statistical measure (e.g., **information gain**)
- Partition tuples recursively based on selected attributes
  - Tree pruning
    - » Identify and remove branches that reflect noise or outliers

## Phase 2 : Use of decision tree: Classifying an unknown sample

- Test the attribute values of the sample against the decision tree

# Entropy & Information Gain (ID3, C4.5)

## Entropy

It is a measure of impurity of an attribute or feature adopted by many algorithms such as ID3 and C5.0.

$$\text{Entropy}(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

$c$  is the number of different class labels

$p$  refers to the proportion of values falling into the  $i$ -th class label

## Information gain

- The information gain is calculated on the basis of the decrease in entropy ( $S$ ) after a data set is split according to a particular attribute ( $A$ ).
- Constructing a decision tree is all about **finding an attribute** that returns the **highest information gain**

Information gain for a particular feature  $A$  is calculated by the difference in entropy before a split with the entropy after the split.

$$\text{Information Gain}(S, A) = \text{Entropy}(S_{bs}) - \text{Entropy}(S_{as})$$

the weighted summation of the entropy for each partition after split

$$\mathbf{Entropy}(S_{as}) = \sum_{i=1}^n w_i \mathbf{Entropy}(p_i)$$

$w_i$  - the proportion of examples falling into each partition

**Q:** Apply the ID3 algorithm to build a decision tree to play tennis or not from the following data . Discuss each step in detail.

| Day | Outlook  | Temperature | Humidity | Wind   | Decision |
|-----|----------|-------------|----------|--------|----------|
| 1   | Sunny    | Hot         | High     | Weak   | No       |
| 2   | Sunny    | Hot         | High     | Strong | No       |
| 3   | Overcast | Hot         | High     | Weak   | Yes      |
| 4   | Rain     | Mild        | High     | Weak   | Yes      |
| 5   | Rain     | Cool        | Normal   | Weak   | Yes      |
| 6   | Rain     | Cool        | Normal   | Strong | No       |
| 7   | Overcast | Cool        | Normal   | Strong | Yes      |
| 8   | Sunny    | Mild        | High     | Weak   | No       |
| 9   | Sunny    | Cool        | Normal   | Weak   | Yes      |
| 10  | Rain     | Mild        | Normal   | Weak   | Yes      |
| 11  | Sunny    | Mild        | Normal   | Strong | Yes      |
| 12  | Overcast | Mild        | High     | Strong | Yes      |
| 13  | Overcast | Hot         | Normal   | Weak   | Yes      |
| 14  | Rain     | Mild        | High     | Strong | No       |

## Target Variable

Decision = {Yes, No}

- Yes = 9
- No = 5

$$\text{Entropy}(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

$$\text{Entropy}(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14}$$

$$\text{Entropy}(S) \approx 0.940$$

| Day | Outlook | Temperature | humidity | Wind   | Decision |
|-----|---------|-------------|----------|--------|----------|
| 1   | Sunny   | Hot         | High     | Weak   | No       |
| 2   | Sunny   | Hot         | High     | Strong | No       |
| 6   | Rain    | Cool        | Normal   | Strong | No       |
| 8   | Sunny   | Mild        | High     | Weak   | No       |
| 14  | Rain    | Mild        | High     | Strong | No       |

| Day | Outlook  | Temperature | humidity | Wind   | Decision |
|-----|----------|-------------|----------|--------|----------|
| 3   | Overcast | Hot         | High     | Weak   | Yes      |
| 4   | Rain     | Mild        | High     | Weak   | Yes      |
| 5   | Rain     | Cool        | Normal   | Weak   | Yes      |
| 7   | Overcast | Cool        | Normal   | Strong | Yes      |
| 9   | Sunny    | Cool        | Normal   | Weak   | Yes      |
| 10  | Rain     | Mild        | Normal   | Weak   | Yes      |
| 11  | Sunny    | Mild        | Normal   | Strong | Yes      |
| 12  | Overcast | Mild        | High     | Strong | Yes      |
| 13  | Overcast | Hot         | Normal   | Weak   | Yes      |

# Calculate Information Gain for Each Attribute

## Outlook

Sunny (5 samples): 2 Yes, 3 No

$$Entropy(Sunny) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.971$$

| Day | Outlook | Temperature | humidity | Wind   | Decision |
|-----|---------|-------------|----------|--------|----------|
| 1   | Sunny   | Hot         | High     | Weak   | No       |
| 2   | Sunny   | Hot         | High     | Strong | No       |
| 8   | Sunny   | Mild        | High     | Weak   | No       |
| 9   | Sunny   | Cool        | Normal   | Weak   | Yes      |
| 11  | Sunny   | Mild        | Normal   | Strong | Yes      |

Overcast (4 samples): 4 Yes, 0 No

$$Entropy(Overcast) = 0$$

| Day | Outlook  | Temperature | humidity | Wind   | Decision |
|-----|----------|-------------|----------|--------|----------|
| 3   | Overcast | Hot         | High     | Weak   | Yes      |
| 7   | Overcast | Cool        | Normal   | Strong | Yes      |
| 12  | Overcast | Mild        | High     | Strong | Yes      |
| 13  | Overcast | Hot         | Normal   | Weak   | Yes      |

Rain (5 samples): 3 Yes, 2 No

$$Entropy(Rain) = 0.971$$

| Day | Outlook | Temperature | humidity | Wind   | Decision |
|-----|---------|-------------|----------|--------|----------|
| 4   | Rain    | Mild        | High     | Weak   | Yes      |
| 5   | Rain    | Cool        | Normal   | Weak   | Yes      |
| 6   | Rain    | Cool        | Normal   | Strong | No       |
| 10  | Rain    | Mild        | Normal   | Weak   | Yes      |
| 14  | Rain    | Mild        | High     | Strong | No       |

$$\text{Entropy}(\text{Sunny}) = 0.971$$

$$\text{Entropy}(\text{Overcast}) = 0$$

$$\text{Entropy}(\text{Rain}) = 0.971$$

$$\text{Information Gain (S, A)} = \text{Entropy (S}_{bs}) - \text{Entropy (S}_{as})$$

$$\text{Entropy (S}_{as}) = \sum_{i=1}^n w_i \text{Entropy (} p_i)$$

$$\text{Gain}(S, \text{Outlook}) = 0.940 - \left( \frac{5}{14} \cdot 0.971 + \frac{4}{14} \cdot 0 + \frac{5}{14} \cdot 0.971 \right)$$

$$\text{Gain}(S, \text{Outlook}) = 0.940 - 0.693 = 0.247$$



## Temperature

- **Hot (4):** 2 Yes, 2 No → Entropy =  $-2/4 * \log 2/4 - 2/4 * \log 2/4 = 1.0$
- **Mild (6):** 4 Yes, 2 No → Entropy =  $-4/6 * \log 4/6 - 2/6 * \log 2/6 = 0.918$
- **Cool (4):** 3 Yes, 1 No → Entropy =  $-3/4 * \log 3/4 - 1/4 * \log 1/4 = 0.811$

| Day | Outlook  | Temperature | humidity | Wind   | Decision |
|-----|----------|-------------|----------|--------|----------|
| 1   | Sunny    | Hot         | High     | Weak   | No       |
| 2   | Sunny    | Hot         | High     | Strong | No       |
| 3   | Overcast | Hot         | High     | Weak   | Yes      |
| 13  | Overcast | Hot         | Normal   | Weak   | Yes      |

| Day | Outlook  | Temperature | humidity | Wind   | Decision |
|-----|----------|-------------|----------|--------|----------|
| 5   | Rain     | Cool        | Normal   | Weak   | Yes      |
| 6   | Rain     | Cool        | Normal   | Strong | No       |
| 7   | Overcast | Cool        | Normal   | Strong | Yes      |
| 9   | Sunny    | Cool        | Normal   | Weak   | Yes      |

| Day | Outlook  | Temperature | humidity | Wind   | Decision |
|-----|----------|-------------|----------|--------|----------|
| 4   | Rain     | Mild        | High     | Weak   | Yes      |
| 8   | Sunny    | Mild        | High     | Weak   | No       |
| 10  | Rain     | Mild        | Normal   | Weak   | Yes      |
| 11  | Sunny    | Mild        | Normal   | Strong | Yes      |
| 12  | Overcast | Mild        | High     | Strong | Yes      |
| 14  | Rain     | Mild        | High     | Strong | No       |

$$Gain(S, Temperature) = 0.940 - \left( \frac{4}{14} \cdot 1.0 + \frac{6}{14} \cdot 0.918 + \frac{4}{14} \cdot 0.811 \right)$$

$$Gain(S, Temperature) = 0.029$$

## Humidity

- **High (7):** 3 Yes, 4 No  $\rightarrow$  Entropy =  $-3/7 \log 3/7 - 4/7 \log 4/7 = 0.985$
- **Normal (7):** 6 Yes, 1 No  $\rightarrow$  Entropy =  $-6/7 \log 6/7 - 1/7 \log 1/7 = 0.591$

| Day | Outlook  | Temperature | humidity | Wind   | Decision |
|-----|----------|-------------|----------|--------|----------|
| 1   | Sunny    | Hot         | High     | Weak   | No       |
| 2   | Sunny    | Hot         | High     | Strong | No       |
| 3   | Overcast | Hot         | High     | Weak   | Yes      |
| 4   | Rain     | Mild        | High     | Weak   | Yes      |
| 8   | Sunny    | Mild        | High     | Weak   | No       |
| 12  | Overcast | Mild        | High     | Strong | Yes      |
| 14  | Rain     | Mild        | High     | Strong | No       |

| Day | Outlook  | Temperature | humidity | Wind   | Decision |
|-----|----------|-------------|----------|--------|----------|
| 5   | Rain     | Cool        | Normal   | Weak   | Yes      |
| 6   | Rain     | Cool        | Normal   | Strong | No       |
| 7   | Overcast | Cool        | Normal   | Strong | Yes      |
| 9   | Sunny    | Cool        | Normal   | Weak   | Yes      |
| 10  | Rain     | Mild        | Normal   | Weak   | Yes      |
| 11  | Sunny    | Mild        | Normal   | Strong | Yes      |
| 13  | Overcast | Hot         | Normal   | Weak   | Yes      |

$$Gain(S, Humidity) = 0.940 - \left( \frac{7}{14} \cdot 0.985 + \frac{7}{14} \cdot 0.591 \right)$$

$$Gain(S, Humidity) = 0.151$$

## Wind

- **Weak (8):** 6 Yes, 2 No → Entropy = 0.811
- **Strong (6):** 3 Yes, 3 No → Entropy = 1.0

| Day | Outlook  | Temperature | humidity | Wind   | Decision |
|-----|----------|-------------|----------|--------|----------|
| 2   | Sunny    | Hot         | High     | Strong | No       |
| 6   | Rain     | Cool        | Normal   | Strong | No       |
| 7   | Overcast | Cool        | Normal   | Strong | Yes      |
| 11  | Sunny    | Mild        | Normal   | Strong | Yes      |
| 12  | Overcast | Mild        | High     | Strong | Yes      |
| 14  | Rain     | Mild        | High     | Strong | No       |

| Day | Outlook  | Temperature | humidity | Wind | Decision |
|-----|----------|-------------|----------|------|----------|
| 1   | Sunny    | Hot         | High     | Weak | No       |
| 3   | Overcast | Hot         | High     | Weak | Yes      |
| 4   | Rain     | Mild        | High     | Weak | Yes      |
| 5   | Rain     | Cool        | Normal   | Weak | Yes      |
| 8   | Sunny    | Mild        | High     | Weak | No       |
| 9   | Sunny    | Cool        | Normal   | Weak | Yes      |
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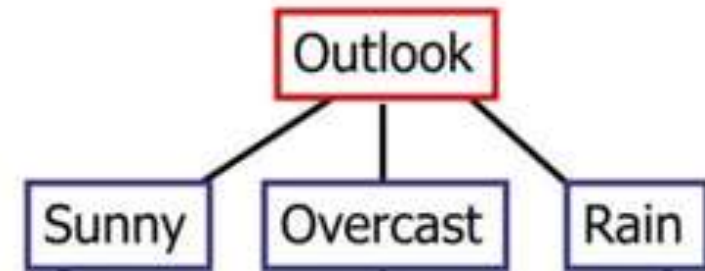
$$Gain(S, Wind) = 0.940 - \left( \frac{8}{14} \cdot 0.811 + \frac{6}{14} \cdot 1.0 \right)$$

$$Gain(S, Wind) = 0.048$$

## Best Attribute

- Outlook: **0.247** (highest)
- Temperature: 0.029
- Humidity: 0.151
- Wind: 0.048

Hence, Root Node = **Outlook**



- Entropy(Sunny)= 0.971
- (Information Gain) for remaining attributes: **Temperature, Humidity, Wind**

- **Temperature within sunny**

- **Hot (2):** 0 Yes, 2 No → Entropy =0
- **Mild (2):** 1 Yes, 1 No →  
Entropy =  $-1/2 \cdot \log 1/2 - 1/2 \cdot \log 1/2 = 1$
- **Cool (1):** 1 Yes, 0 No → Entropy = 0

| Day | Outlook | Temperature | humidity | Wind   | Decision |
|-----|---------|-------------|----------|--------|----------|
| 1   | Sunny   | Hot         | High     | Weak   | No       |
| 2   | Sunny   | Hot         | High     | Strong | No       |
| 8   | Sunny   | Mild        | High     | Weak   | No       |
| 9   | Sunny   | Cool        | Normal   | Weak   | Yes      |
| 11  | Sunny   | Mild        | Normal   | Strong | Yes      |

$$E_{Temp} = \frac{2}{5} \cdot 0 + \frac{2}{5} \cdot 1.0 + \frac{1}{5} \cdot 0 = 0.4$$

$$Gain(S, Temp) = 0.97095 - 0.4 = 0.57095$$

- **Humidity** within sunny

- **High (3)**: 0 Yes, 3 No → Entropy = 0
- **Normal (2)**: 2 Yes, 0 No → Entropy = 0

$$Gain(S, Humidity) = 0.97095 - 0 = 0.97095$$

## Wind

**Weak (3)**: 1 Yes, 2 No → Entropy = 0.918

**Strong (2)**: 1 Yes, 1 No → Entropy = 1.0

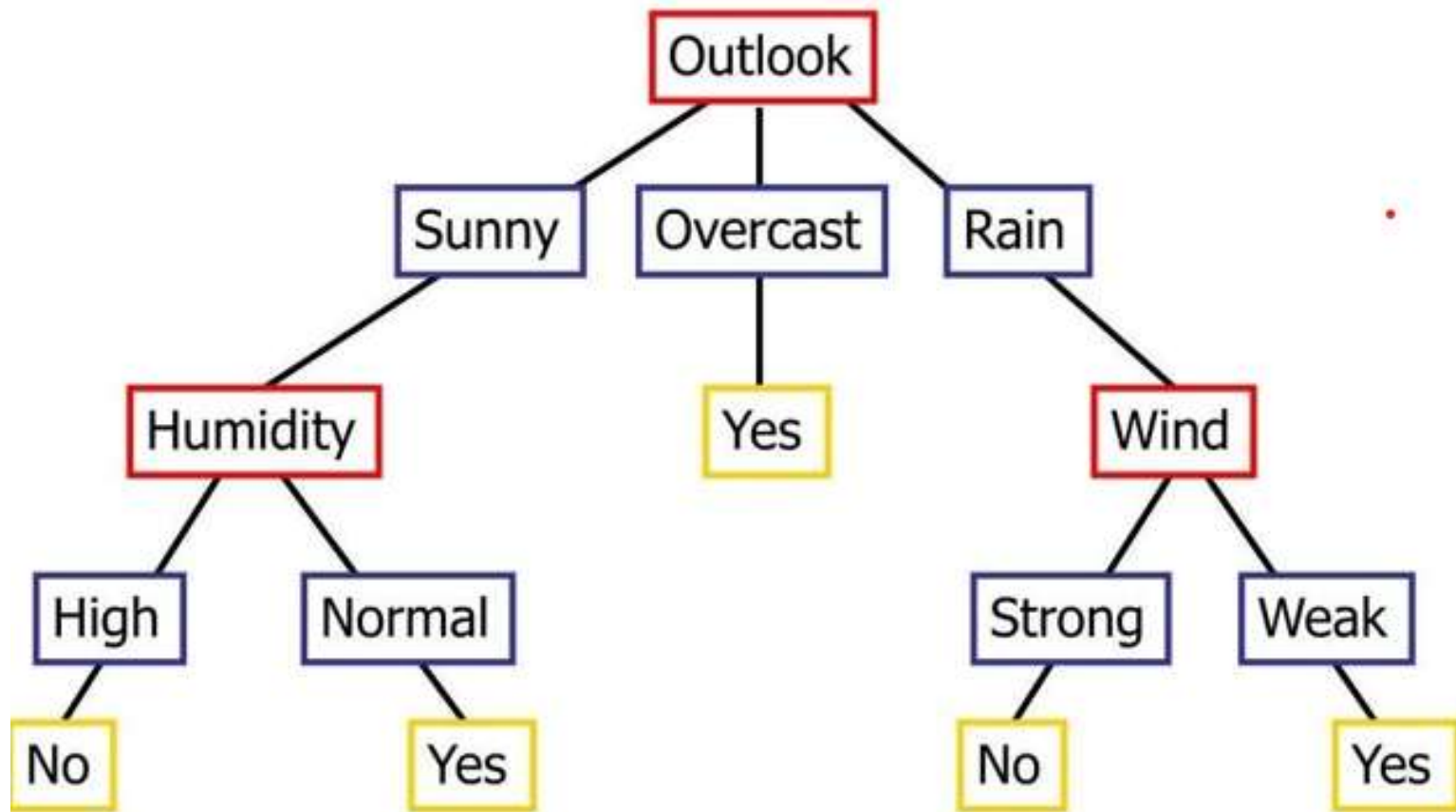
$$E_{Wind} = \frac{3}{5} \cdot 0.91830 + \frac{2}{5} \cdot 1.0 \approx 0.95098$$

$$Gain(S, Wind) = 0.97095 - 0.95098 \approx 0.01997$$

| Day | Outlook | Temperature | humidity | Wind   | Decision |
|-----|---------|-------------|----------|--------|----------|
| 1   | Sunny   | Hot         | High     | Weak   | No       |
| 2   | Sunny   | Hot         | High     | Strong | No       |
| 8   | Sunny   | Mild        | High     | Weak   | No       |
| 9   | Sunny   | Cool        | Normal   | Weak   | Yes      |
| 11  | Sunny   | Mild        | Normal   | Strong | Yes      |

- Choose best attribute for level 2 inside Sunny
- Compare gains:
  - $\text{Gain}(\text{Temperature}) \approx 0.57095$
  - $\text{Gain}(\text{Humidity}) \approx 0.97095 \leftarrow \text{largest}$
  - $\text{Gain}(\text{Wind}) \approx 0.01997$
- Best split = Humidity.

Similarly calculate for Rain





# Computing Information-Gain for Continuous-Valued Attributes

- Let attribute  $A$  be a continuous-valued attribute
- Must determine the *best split point* for  $A$ 
  - Sort the value  $A$  in increasing order
  - Typically, the midpoint between each pair of adjacent values is considered as a possible *split point*
    - $(a_i + a_{i+1})/2$  is the midpoint between the values of  $a_i$  and  $a_{i+1}$
  - The point with the *minimum expected information requirement* for  $A$  is selected as the split-point for  $A$
- Split:
  - $D_1$  is the set of tuples in  $D$  satisfying  $A \leq \text{split-point}$ , and  $D_2$  is the set of tuples in  $D$  satisfying  $A > \text{split-point}$

# Gini Index (CART)

- If a data set  $D$  contains examples from  $n$  classes, gini index,  $gini(D)$  is defined as

$$gini(D) = 1 - \sum_{j=1}^n p_j^2$$

where  $p_j$  is the relative frequency of class  $j$  in  $D$

- If a data set  $D$  is split on  $A$  into two subsets  $D_1$  and  $D_2$ , the *gini* index  $gini(D)$  or weighted gini is defined as

$$gini_A(D) = \frac{|D_1|}{|D|} gini(D_1) + \frac{|D_2|}{|D|} gini(D_2)$$

- Reduction in Impurity:

$$\Delta gini(A) = gini(D) - gini_A(D)$$

- The attribute provides the smallest  $gini_{split}(D)$  (or the largest reduction in impurity) is chosen to split the node (*need to enumerate all the possible splitting points for each attribute*)

# Example: Play Tennis

Total = 14 samples: Yes = 9, No = 5

$$Gini(S) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 1 - \frac{81}{196} - \frac{25}{196} = 0.459$$

## (A) Outlook

- Sunny (5) → Yes=2, No=3 →  $Gini = 0.48 = 1 - (2/5 * 2/5) - (3/5 * 3/5)$
- Overcast (4) → Yes=4, No=0 →  $Gini = 0.0$
- Rain (5) → Yes=3, No=2 →  $Gini = 0.48$

Weighted:

$$\frac{5}{14}(0.48) + \frac{4}{14}(0) + \frac{5}{14}(0.48) = 0.343$$

$$\text{Gain} = 0.459 - 0.343 = 0.116$$

### (B) Temperature

- Hot (4)  $\rightarrow$  Yes=2, No=2  $\rightarrow$   $Gini = 0.5$
- Mild (6)  $\rightarrow$  Yes=4, No=2  $\rightarrow$   $Gini = 0.444$
- Cool (4)  $\rightarrow$  Yes=3, No=1  $\rightarrow$   $Gini = 0.375$

Weighted:

$$\frac{4}{14}(0.5) + \frac{6}{14}(0.444) + \frac{4}{14}(0.375) = 0.440$$

$$\text{Gain} = 0.459 - 0.440 = 0.019$$

### (C) Humidity

- High (7)  $\rightarrow$  Yes=3, No=4  $\rightarrow$   $Gini = 0.490$
- Normal (7)  $\rightarrow$  Yes=6, No=1  $\rightarrow$   $Gini = 0.245$

Weighted:

$$\frac{7}{14}(0.490) + \frac{7}{14}(0.245) = 0.367$$

$$\text{Gain} = 0.459 - 0.367 = 0.092$$

#### (D) Wind

- Weak (8) → Yes=6, No=2 →  $Gini = 0.375$
- Strong (6) → Yes=3, No=3 →  $Gini = 0.5$

Weighted:

$$\frac{8}{14}(0.375) + \frac{6}{14}(0.5) = 0.429$$

$$\text{Gain} = 0.459 - 0.429 = 0.031$$

**Best Split is the attribute with minimum Gini Index**

**Outlook:** 0.343(**Lowest**) **ROOT**

Temperature: 0.440

Humidity: 0.367

Wind: 0.429

# Outlook = Sunny

| Day | Outlook | Temperature | humidity | Wind   | Decision |
|-----|---------|-------------|----------|--------|----------|
| 1   | Sunny   | Hot         | High     | Weak   | No       |
| 2   | Sunny   | Hot         | High     | Strong | No       |
| 8   | Sunny   | Mild        | High     | Weak   | No       |
| 9   | Sunny   | Cool        | Normal   | Weak   | Yes      |
| 11  | Sunny   | Mild        | Normal   | Strong | Yes      |

## Check Temperature

Sunny rows with Temperature:

- Hot → 2 records (both No) → Gini = 0
- Mild → 2 records (Yes=1, No=1) → Gini = 0.5
- Cool → 1 record (Yes=1) → Gini = 0

Weighted Gini =  $(2/5)*0 + (2/5)*0.5 + (1/5)*0 = 0.2$

## Check Humidity

- Sunny & High → 3 records (all No) → Gini = 0
- Sunny & Normal → 2 records (all Yes) → Gini = 0

Weighted Gini =  $(3/5)*0 + (2/5)*0 = 0.0$

## Check Wind

Sunny rows with Wind:

- Weak → 2 records (Yes=1, No=1) → Gini = 0.5
- Strong → 3 records (Yes=1, No=2) →

$$1 - \left(1/3\right)^2 - \left(2/3\right)^2 = 1 - (1/9 + 4/9) = 4/9 = 0.444$$

Weighted Gini =  $(2/5)*0.5 + (3/5)*0.444 \approx 0.467$

**Humidity has the lowest Gini index. So we pick Humidity as the splitting attribute under Sunny.**

- So **Level 2** means:
- For **Sunny branch**, we split again by **Humidity**.
- For **Rain branch**, we split again by **Wind**.
- For **Overcast branch**, no need (already pure).

# EXAMPLE

- Global Technology Solutions (GTS), a leading provider of IT solutions, is coming to College of Engineering and Management (CEM) for hiring B.Tech. students. Last year during campus recruitment, they had shortlisted 18 students for the final interview.
  - A student of CEM, wants to find out if he may be offered a job in GTS. His CGPA is quite high. His self-evaluation on the other parameters is as follows:
    - *Communication – Bad; Aptitude – High; Programming skills – Bad*



# Example contd...Training data for GTS recruitment

| <b>CGPA</b> | <b>Communication</b> | <b>Aptitude</b> | <b>Programming Skill</b> | <b>Job offered?</b> |
|-------------|----------------------|-----------------|--------------------------|---------------------|
| High        | Good                 | High            | Good                     | Yes                 |
| Medium      | Good                 | High            | Good                     | Yes                 |
| Low         | Bad                  | Low             | Good                     | No                  |
| Low         | Good                 | Low             | Bad                      | No                  |
| High        | Good                 | High            | Bad                      | Yes                 |
| High        | Good                 | High            | Good                     | Yes                 |
| Medium      | Bad                  | Low             | Bad                      | No                  |
| Medium      | Bad                  | Low             | Good                     | No                  |
| High        | Bad                  | High            | Good                     | Yes                 |
| Medium      | Good                 | High            | Good                     | Yes                 |
| Low         | Bad                  | High            | Bad                      | No                  |
| Low         | Bad                  | High            | Bad                      | No                  |
| Medium      | Good                 | High            | Bad                      | Yes                 |
| Low         | Good                 | Low             | Good                     | No                  |
| High        | Bad                  | Low             | Bad                      | No                  |
| Medium      | Bad                  | High            | Good                     | No                  |
| High        | Bad                  | Low             | Bad                      | No                  |
| Medium      | Good                 | High            | Bad                      | Yes                 |

# Ending Tree Growth

- Grow the tree until
  - additional splitting produces no significant information gain
  - statistical test - a chi-squared test
  - problem - trees that are too small
  - only compares one split with the next descending split

# Avoid Over fitting in Classification

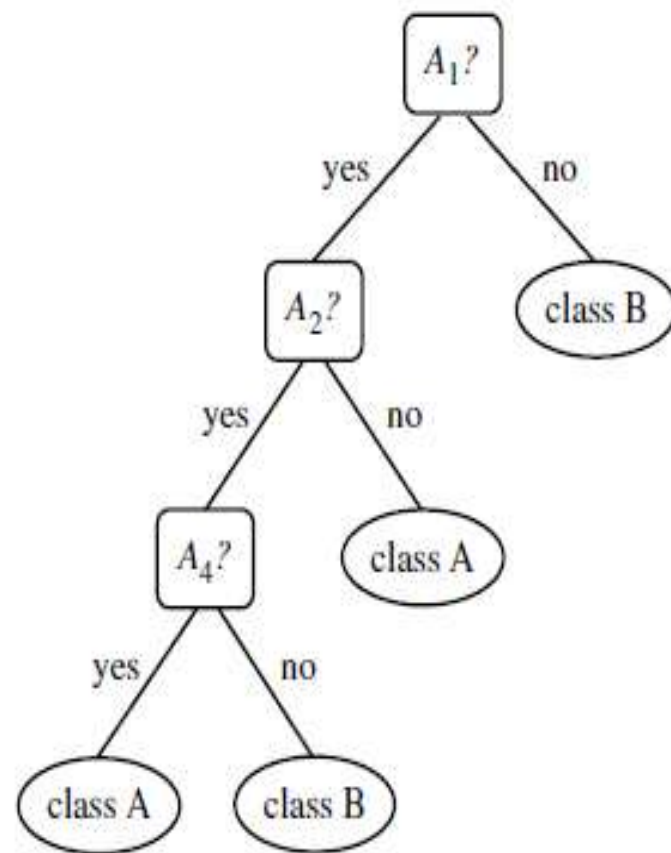
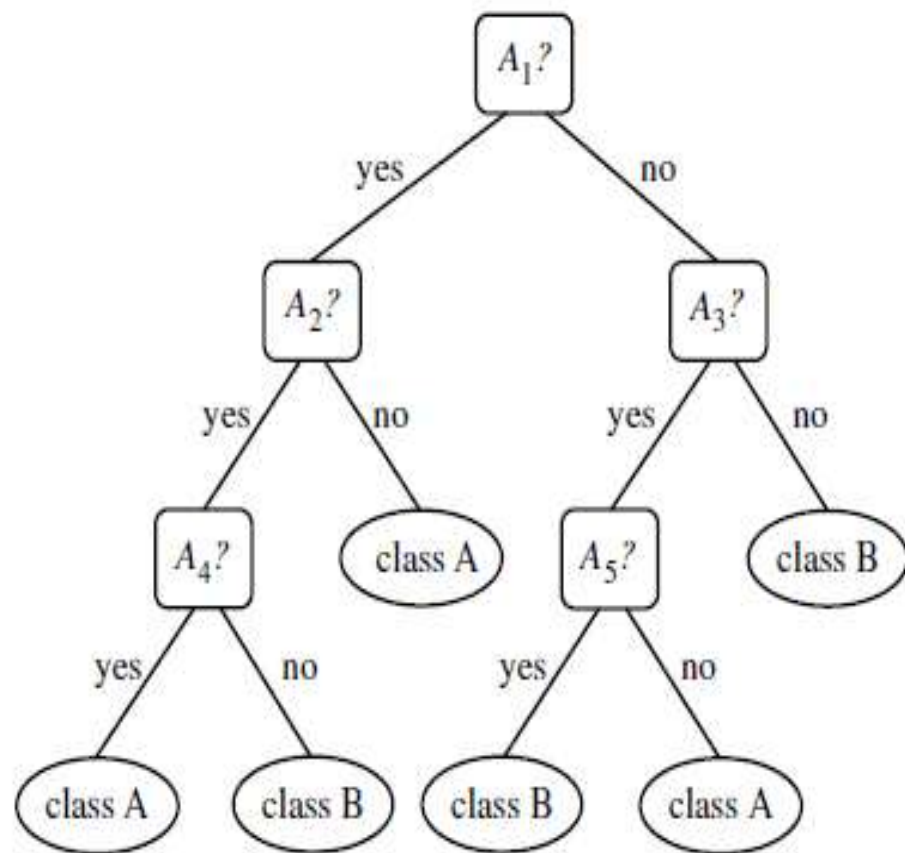
- The generated tree may over-fit the training data
  - Too many branches, some may reflect anomalies due to noise or outliers
  - Use statistical measures to remove the least reliable branches
  - Two approaches to avoid over fitting
- Reduce large tree
  - Reduce its size by eliminating or pruning weak branches step by step
  - Continue until minimum true error rate

# Tree Pruning

1. Pre pruning: Halt tree construction early—do not split a node if this would result in the goodness measure falling below a threshold
  - Difficult to choose an appropriate threshold
2. Post pruning: Remove branches from a “fully grown” tree
  - The leaf is labeled with the most frequent class among the sub-tree being replaced

# Pre pruning

- The leaf may hold the most frequent class among the subset tuples or the probability distribution of those tuples.
- There are difficulties, however, in choosing an appropriate threshold.
  - High thresholds could result in oversimplified trees, whereas low thresholds could result in very little simplification.



# Post Pruning

- Examples of Post Pruning
  - Cost complexity in CART algorithm
    - Function of the number of leaves in the tree and the error rate of the tree (percentage of tuples misclassified by the tree)
- Pruning set of class-labeled tuples is used to estimate cost complexity.
- This set is independent of the training set used to build the unpruned tree and of any test set used for accuracy estimation

- C4.5 uses a method called pessimistic pruning, which is similar to the cost complexity method in that it also uses error rate estimates to make decisions regarding sub-tree pruning.
- Pessimistic pruning, however, does not require the use of a prune set
- it uses the training set to estimate error rates



# Naïve Bayes Classification

# Naïve Bayesian Classification

From Bayes Theorem

$$P(C_i | X) = \frac{P(X | C_i)P(C_i)}{P(X)}$$

$P(X)$  is constant. Only  $P(X | C_i)P(C_i)$  need be maximized.

- If class prior probabilities not known, then assume all classes to be equally likely and therefore maximize  $P(X|C_i)$
- Otherwise maximize

$$P(X | C_i)P(C_i)$$

# Naïve Bayesian Classification

4. Naïve assumption: attribute independence *class conditional independence*

$$4. P(X|C_i) = P(x_1, \dots, x_n|C) = \prod_{k=1}^n P(X_k|C_i)$$

5. In order to classify an unknown sample  $X$ , evaluate  $P(X|C_i)P(C_i)$  for each class  $C_i$ . Sample  $X$  is assigned to the class  $C_i$  if

$$P(X|C_i)P(C_i) > P(X|C_j)P(C_j) \text{ for } 1 \leq j \leq m, j \neq i$$

# Naïve Bayes Classifier: Training Dataset

Class:

C1:buys\_computer = 'yes'

C2:buys\_computer = 'no'

Data to be classified:

X = (age <=30,

Income = medium,

Student = yes

Credit\_rating = Fair)

| age     | income | student | credit_rating | com |
|---------|--------|---------|---------------|-----|
| <=30    | high   | no      | fair          | no  |
| <=30    | high   | no      | excellent     | no  |
| 31...40 | high   | no      | fair          | yes |
| >40     | medium | no      | fair          | yes |
| >40     | low    | yes     | fair          | yes |
| >40     | low    | yes     | excellent     | no  |
| 31...40 | low    | yes     | excellent     | yes |
| <=30    | medium | no      | fair          | no  |
| <=30    | low    | yes     | fair          | yes |
| >40     | medium | yes     | fair          | yes |
| <=30    | medium | yes     | excellent     | yes |
| 31...40 | medium | no      | excellent     | yes |
| 31...40 | high   | yes     | fair          | yes |
| >40     | medium | no      | excellent     | no  |

# Naïve Bayes Classifier: An Example

| age     | income | student | credit_rating | com |
|---------|--------|---------|---------------|-----|
| <=30    | high   | no      | fair          | no  |
| <=30    | high   | no      | excellent     | no  |
| 31...40 | high   | no      | fair          | yes |
| >40     | medium | no      | fair          | yes |
| >40     | low    | yes     | fair          | yes |
| >40     | low    | yes     | excellent     | no  |
| 31...40 | low    | yes     | excellent     | yes |
| <=30    | medium | no      | fair          | no  |
| <=30    | low    | yes     | fair          | yes |
| >40     | medium | yes     | fair          | yes |
| <=30    | medium | yes     | excellent     | yes |
| 31...40 | medium | no      | excellent     | yes |
| 31...40 | high   | yes     | fair          | yes |
| >40     | medium | no      | excellent     | no  |

- $P(C_i)$ :  $P(\text{buys\_computer} = \text{"yes"}) = 9/14 = 0.643$

$$P(\text{buys\_computer} = \text{"no"}) = 5/14 = 0.357$$

- Compute  $P(X|C_i)$  for each class

$$P(\text{age} = \text{"<=30"} \mid \text{buys\_computer} = \text{"yes"}) = 2/9 = 0.222$$

$$P(\text{age} = \text{"<= 30"} \mid \text{buys\_computer} = \text{"no"}) = 3/5 = 0.6$$

$$P(\text{income} = \text{"medium"} \mid \text{buys\_computer} = \text{"yes"}) = 4/9 = 0.444$$

$$P(\text{income} = \text{"medium"} \mid \text{buys\_computer} = \text{"no"}) = 2/5 = 0.4$$

$$P(\text{student} = \text{"yes"} \mid \text{buys\_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{student} = \text{"yes"} \mid \text{buys\_computer} = \text{"no"}) = 1/5 = 0.2$$

$$P(\text{credit\_rating} = \text{"fair"} \mid \text{buys\_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{credit\_rating} = \text{"fair"} \mid \text{buys\_computer} = \text{"no"}) = 2/5 = 0.4$$

- **$X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit\_rating} = \text{fair})$**

$$P(X|C_i) : P(X \mid \text{buys\_computer} = \text{"yes"}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = 0.044$$

$$P(X \mid \text{buys\_computer} = \text{"no"}) = 0.6 \times 0.4 \times 0.2 \times 0.4 = 0.019$$

$$P(X|C_i) * P(C_i) : P(X \mid \text{buys\_computer} = \text{"yes"}) * P(\text{buys\_computer} = \text{"yes"}) = 0.028$$

$$P(X \mid \text{buys\_computer} = \text{"no"}) * P(\text{buys\_computer} = \text{"no"}) = 0.007$$

**Therefore, X belongs to class ("buys\_computer = yes")**

# Naïve Bayes Classifier: Comments

- Advantages
  - Easy to implement
  - Good results obtained in most of the cases
- Disadvantages
  - Assumption: class conditional independence, therefore loss of accuracy
  - Practically, dependencies exist among variables
    - E.g., hospitals: patients: Profile: age, family history, etc.  
Symptoms: fever, cough etc., Disease: lung cancer, diabetes, etc.
    - Dependencies among these cannot be modeled by Naïve Bayes Classifier
- How to deal with these dependencies? Bayesian Belief Networks

# Bayesian Belief Networks

# Difference between Naïve Bayes and BBN

| Naïve Bayes Classifier (NB)                                         | Bayesian Belief Networks (BBN)                                                                     |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Assumes <b>conditional independence</b> of features given the class | Can represent <b>complex dependencies</b> between variables                                        |
| Uses class prior + feature likelihoods                              | Each node has a Conditional Probability Table (CPT)                                                |
| A simple <b>star-shaped graph</b> : one class node → all features   | Directed Acyclic Graph ( <b>DAG</b> )                                                              |
| Very fast and simple (direct multiplication of probabilities)       | More computationally expensive (requires inference algorithms like variable elimination, sampling) |



# Bayesian Belief Networks

- It is a **probabilistic graphical model**.
- It represents a set of **variables** and their **conditional dependencies** using a **directed acyclic graph (DAG)**.
- Requires that graph is acyclic (no directed cycles)

## Components

- **Nodes** → Represent random variables (e.g., Weather, Disease, Exam).
- **Edges** → Directed arrows showing probabilistic dependency (e.g., Rain → WetGround).
- **CPT (Conditional Probability Table)** → Each node has a **Conditional Probability Table (CPT)** that defines the probability of the variable given its parents.

# Bayesian Belief Networks

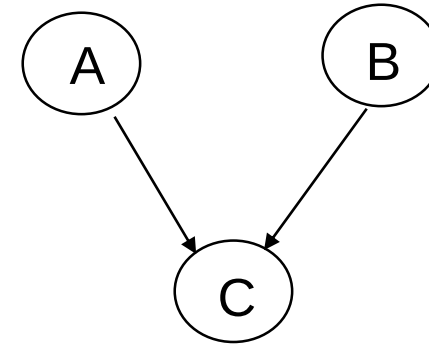
For a Bayesian Belief Network, the joint probability of all variables  $X_1, X_2, \dots, X_n$  is:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Parents}(X_i))$$

# Example of a simple Bayesian network

$$P(X_1, X_2, \dots, X_N) = \prod_i P(X_i \mid \text{parents}(X_i))$$

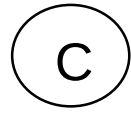
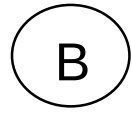
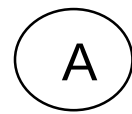
$$P(A, B, C) = P(C|A, B)P(A)P(B) \longleftrightarrow$$



Probability model has simple factored form

- Directed edges => direct dependence
- Absence of an edge => conditional independence
- Also known as belief networks, graphical models, causal networks

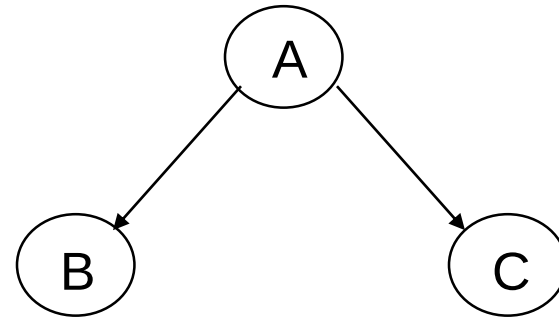
# Examples of 3-way Bayesian Networks



Absolute Independence:  
 $p(A,B,C) = p(A) p(B) p(C)$

# Examples of 3-way Bayesian Networks

- Conditionally independent effects:  
$$p(A, B, C) = p(B|A)p(C|A)p(A)$$
- B and C are conditionally independent given A
- e.g., A is a disease, and we model B and C as conditionally independent symptoms given A

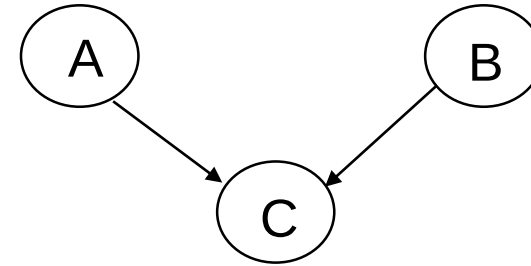


# Examples of 3-way Bayesian Networks

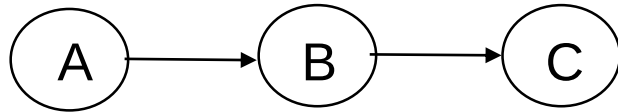
- Independent Clauses:

$$p(A, B, C) = p(C|A, B)p(A)p(B)$$

- “Explaining away” effect:
  - A and B are independent but become dependent once C is known!!



# Examples of 3-way Bayesian Networks



Markov dependence:

$$p(A,B,C) = p(C|B) p(B|A)p(A)$$

# The Alarm Example

- You have a new burglar alarm installed
- It is reliable about detecting burglary, but responds to minor earthquakes
- Two neighbors (John, Mary) promise to call you at work when they hear the alarm
  - John always calls when hears alarm, but confuses alarm with phone ringing (and calls then also)
  - Mary likes loud music and sometimes misses alarm!
- Given evidence about who has and hasn't called, estimate the probability of a burglary



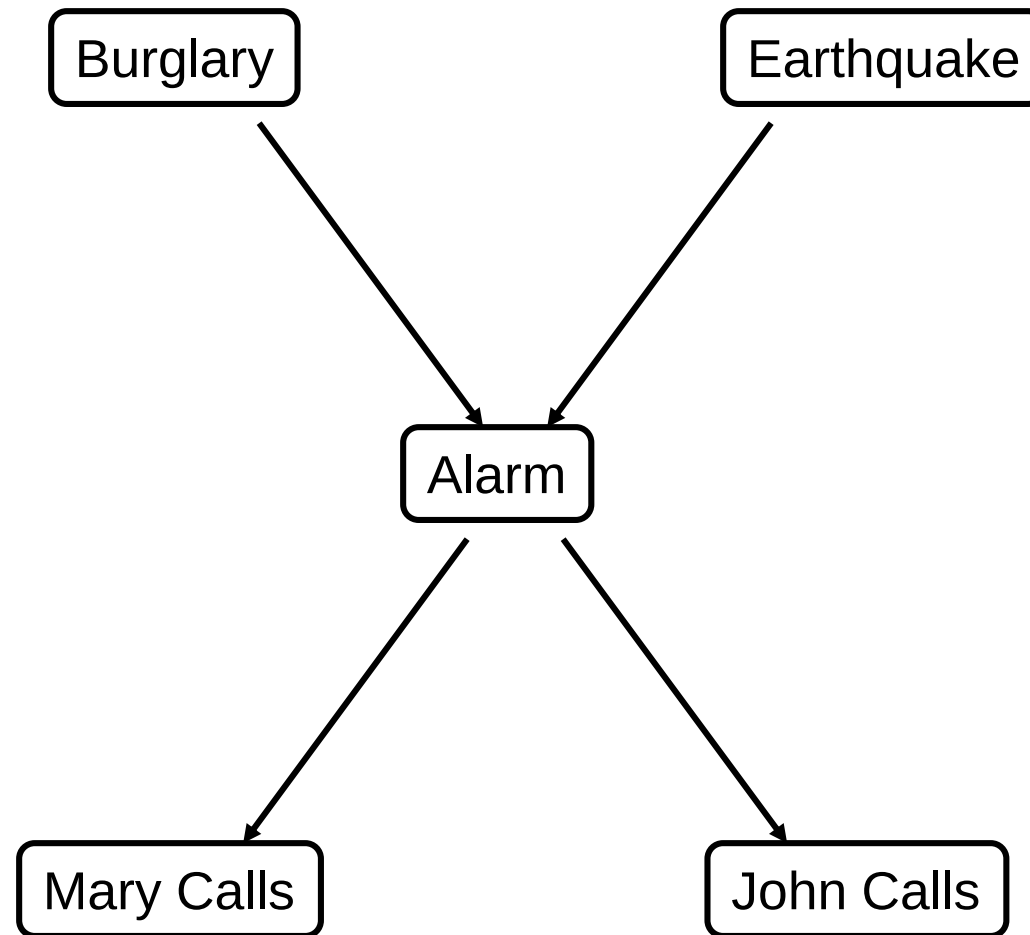
# The Alarm Example

- Represent problem using 5 binary variables:
  - B = a burglary occurs at your house
  - E = an earthquake occurs at your house
  - A = the alarm goes off
  - J = John calls to report the alarm
  - M = Mary calls to report the alarm
- What is  $P(B \mid M, J)$  ?
  - We can use the full joint distribution to answer this question
    - Requires  $2^5 = 32$  probabilities
  - Can we use prior domain knowledge to come up with a Bayesian network that requires fewer probabilities?

# Constructing a Bayesian Network: Step 1

- Order the variables in terms of causality (may be a partial order)
  - e.g.,  $\{E, B\} \rightarrow \{A\} \rightarrow \{J, M\}$
- Use these assumptions to create the graph structure of the Bayesian network

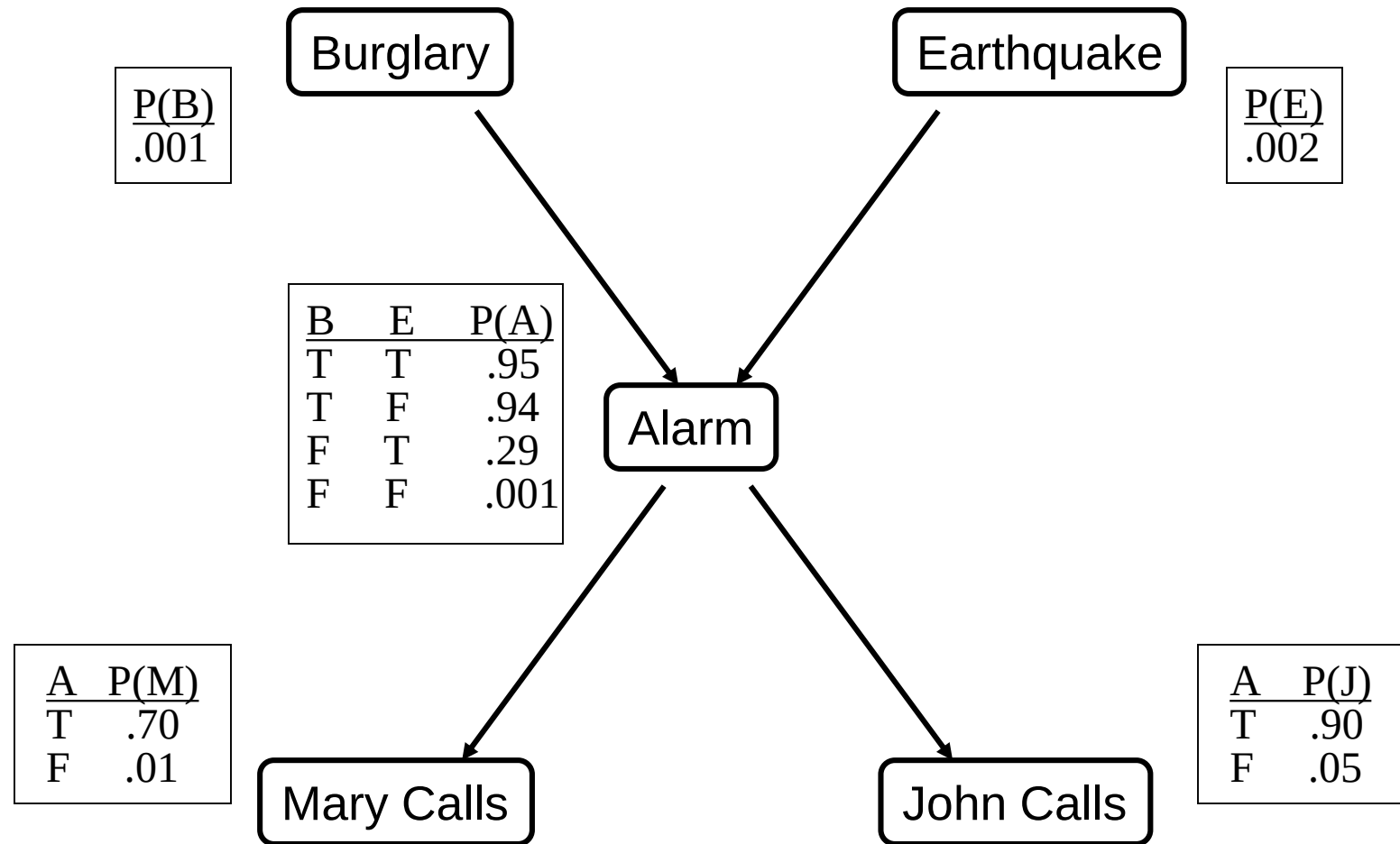
# Earthquake or Burglar



# Constructing a Bayesian Network: Step 2

- Fill in conditional probability tables (CPTs)
  - One for each node
  - $2^p$  entries, where  $p$  is the number of parents
- Where do these probabilities come from?
  - Expert knowledge
  - From data (relative frequency estimates)
  - Or a combination of both

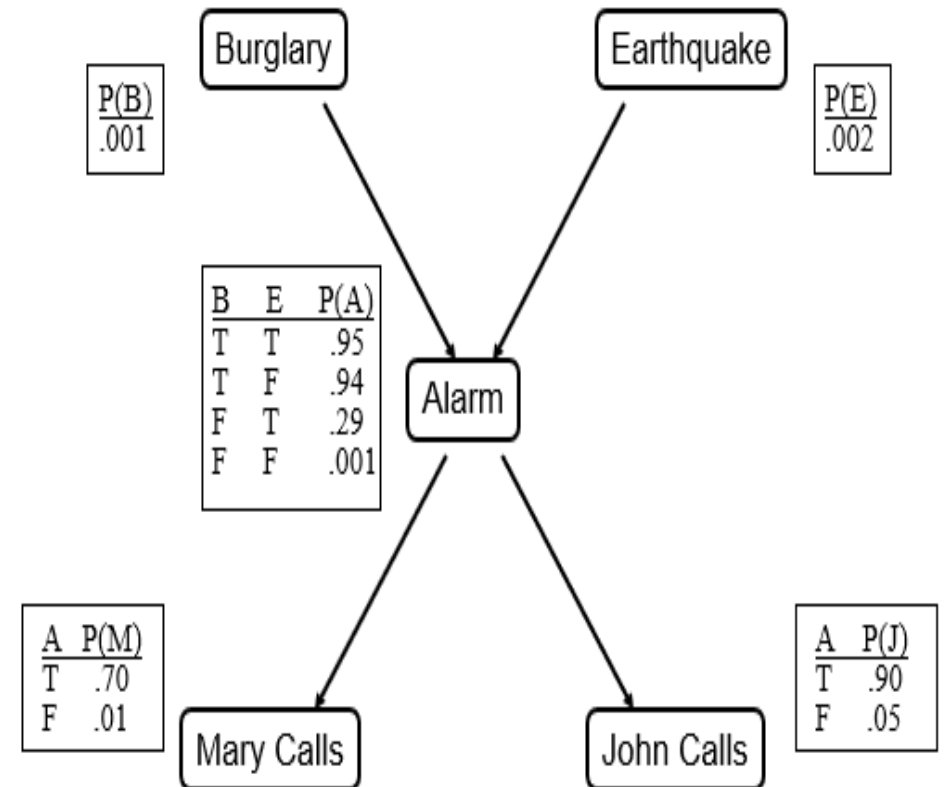
# Bayesian Network for Alarm Domain



# Example

Q: What is the probability that the alarm has sounded but neither a burglary nor an earthquake has occurred and both John and Merry called.

$$\begin{aligned} & P(J \& M \& A \& \neg B \& \neg E) \\ = & P(J|A)P(M|A)P(A|\neg B, \neg E)P(\neg B)P(\neg E) \\ = & 0.9 \times 0.7 \times 0.001 \times 0.999 \times 0.998 \\ = & 0.00062 \end{aligned}$$

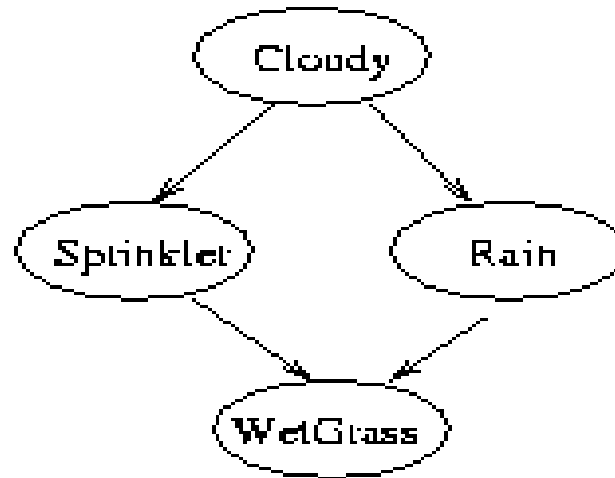


Q: what is the probability that a burglary occurs, no earthquake happens, the alarm rings, Mary calls, and John does not call

$$\begin{aligned} P(\neg j, m, a, \neg e, b) &\approx P(\neg j \mid a) P(m \mid a) P(a \mid \neg e, b) P(\neg e) P(b) \\ &\approx 0.10 \times 0.70 \times 0.94 \times 0.998 \times 0.001 \\ &\approx 0.00006566 \end{aligned}$$

# BAYESIAN BELIEF NETWORK Example-2

| $P(C=F)$ | $P(C=T)$ |
|----------|----------|
| 0.5      | 0.5      |



| C | $P(S=F)$ | $P(S=T)$ |
|---|----------|----------|
| F | 0.5      | 0.5      |
| T | 0.9      | 0.1      |

| C | $P(R=F)$ | $P(R=T)$ |
|---|----------|----------|
| F | 0.8      | 0.2      |
| T | 0.2      | 0.8      |

| S | R | $P(W=F)$ | $P(W=T)$ |
|---|---|----------|----------|
| F | F | 1.0      | 0.0      |
| T | F | 0.1      | 0.9      |
| F | T | 0.1      | 0.9      |
| T | T | 0.01     | 0.99     |



# BAYESIAN BELIEF NETWORK

- By the chaining rule of probability, the joint probability of all the nodes in the graph above is:

$$P(C, S, R, W) = P(C) * P(S|C) * P(R|C) * P(W|S,R)$$

W=Wet Grass, C=Cloudy, R=Rain, S=Sprinkler

Example:  $P(W \cap \neg R \cap S \cap C)$

$$= P(W|S, \neg R) * P(\neg R|C) * P(S|C) * P(C)$$

$$= 0.9 * 0.2 * 0.1 * 0.5 = 0.009$$

# BAYESIAN BELIEF NETWORK

What is the probability of wet grass on a given day -  $P(W)$ ?

$$\begin{aligned} P(W) = & P(W|SR) * P(S) * P(R) + \\ & P(W|S-R) * P(S) * P(-R) + \\ & P(W|-SR) * P(-S) * P(R) + \\ & P(W|-S-R) * P(-S) * P(-R) \end{aligned}$$

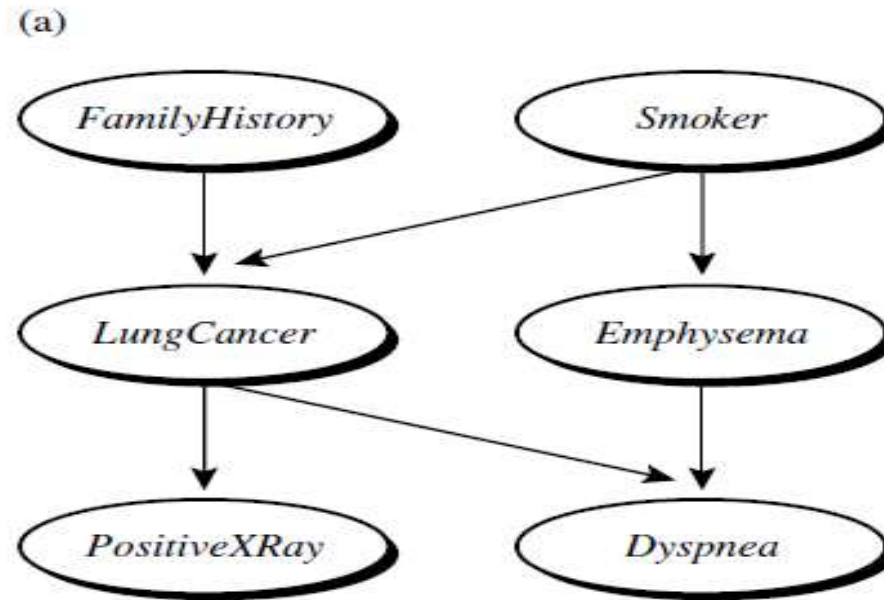
$$\text{Here } P(S) = P(S|C) * P(C) + P(S|-C) * P(-C)$$

$$P(R) = P(R|C) * P(C) + P(R|-C) * P(-C)$$

$$P(W) = 0.5985$$

# Bayesian Belief Networks – EXAMPLE-3

- 6 Boolean variables
- Arcs allow representation of causal knowledge
- Having lung cancer is influenced by family history and smoking
- Positive Xray is ind. of whether the patient has a FH or if he/she is a smoker given that we know that the patient has lung cancer
- Once we know the outcome of Lung Cancer, FH & Smoker do not provide any additional info. about PositiveXray



(b)

|            | <i>FH, S</i> | <i>FH, ~S</i> | <i>~FH, S</i> | <i>~FH, ~S</i> |
|------------|--------------|---------------|---------------|----------------|
| <i>LC</i>  | 0.8          | 0.5           | 0.7           | 0.1            |
| <i>~LC</i> | 0.2          | 0.5           | 0.3           | 0.9            |

Conditional Probability Table for LungCancer

# Bayesian Belief Networks

- Lung Cancer is CI of Emphysema, given its parents, FH & Smoker
- BBN has a Conditional Probability Table (CPT) for each variable
- CPT for a variable Y specifies the conditional distribution  $P(Y|\text{parents}(Y))$

$$P(\text{LC}=\text{Y}|\text{FH}=\text{Y},\text{S}=\text{Y}) = 0.8$$

$$P(\text{LC}=\text{N}|\text{FH}=\text{N},\text{S}=\text{N}) = 0.9$$

|     | (FH, S) | (FH, ~S) | (~FH, S) | (~FH, ~S) |
|-----|---------|----------|----------|-----------|
| LC  | 0.8     | 0.5      | 0.7      | 0.1       |
| ~LC | 0.2     | 0.5      | 0.3      | 0.9       |

CPT for LungCancer

# Advantages of Bayesian Approach

- Bayesian networks can readily handle incomplete data sets.
- Bayesian networks allow one to learn about causal relationships
- Bayesian networks readily facilitate use of prior knowledge.

# SUPPORT VECTOR MACHINE (SVM)

# What is support vector?

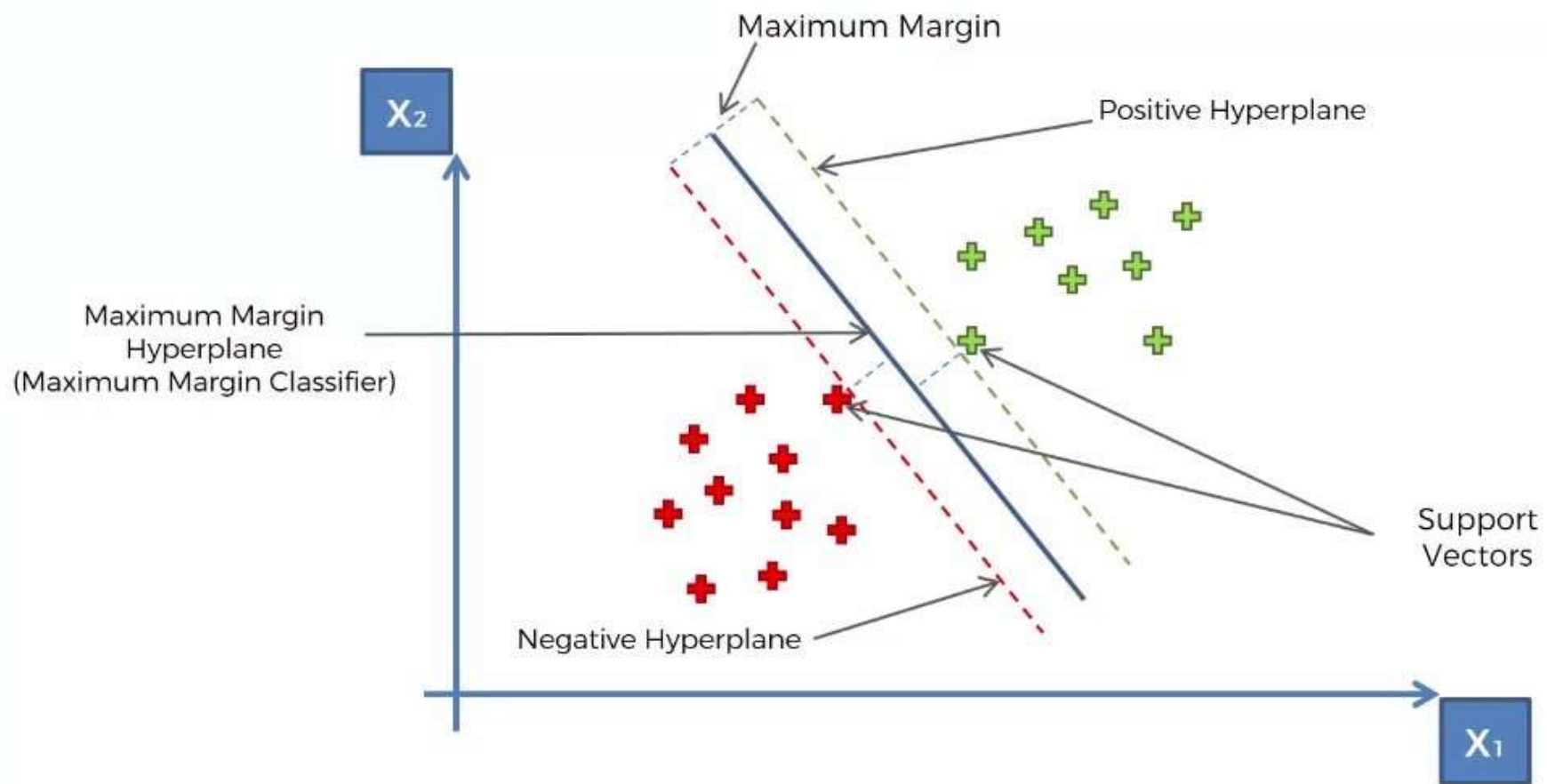
- “Support Vector Machine” (SVM) is a supervised machine learning algorithm which can be used for both classification or regression challenges. However, it is mostly used in classification problems.
- In this algorithm, we plot each data item as a point in  $n$ -dimensional space (where  $n$  is number of features you have) with the value of each feature being the value of a particular coordinate.
- Then, we perform classification by finding the hyperplane that differentiate the two classes very well.

# Support Vector Machine

- Generally, Support Vector Machines is considered to be a classification approach, but can be employed in both types of classification and regression problems.
- It can easily handle multiple continuous and categorical variables.
- SVM constructs a hyperplane in multidimensional space to separate different classes. SVM generates optimal hyperplane in an iterative manner, which is used to minimize an error.
- The core idea of SVM is to find a maximum marginal hyperplane(MMH) that best divides the dataset into classes.



# Decision Vectors



# Definitions

- Support Vectors
  - Support vectors are the **data points**, which are **closest to the hyperplane**. These points will define the separating line better by calculating margins. These points are more relevant to the construction of the classifier.
- Hyperplane
  - A hyperplane is a decision plane which separates between a set of objects having different class memberships.

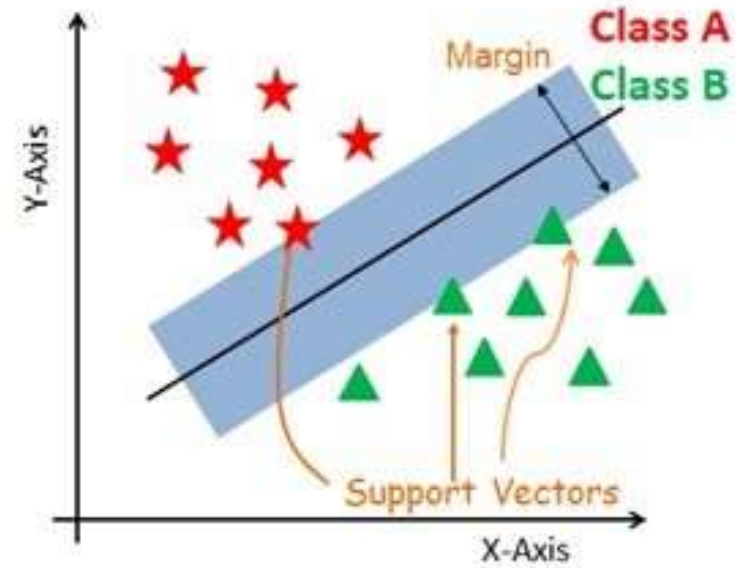
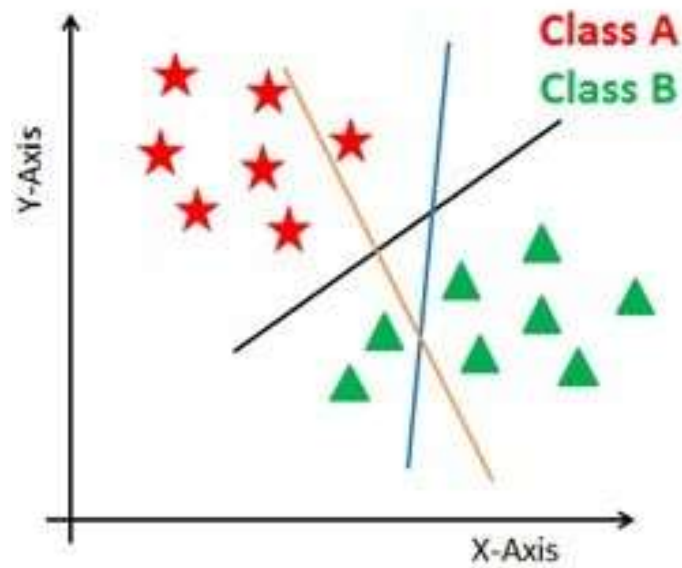
# Definitions

- Margin
  - A margin is a **gap** between the two lines on the closest class points.
  - This is calculated as the perpendicular distance from the line to support vectors or closest points.
  - If the margin is larger in between the classes, then it is considered a good margin, a smaller margin is a bad margin.

# How SVM works ?

- The main objective is to segregate the given dataset in the best possible way.
- The distance between the either nearest points is known as the margin.
- The **objective** is to **select a hyperplane with the maximum possible margin between support vectors** in the given dataset. SVM searches for the maximum marginal hyperplane in the following steps:
  - Generate hyperplanes which segregates the classes in the best way.
  - Select the right hyperplane with the maximum segregation from the either nearest data points.

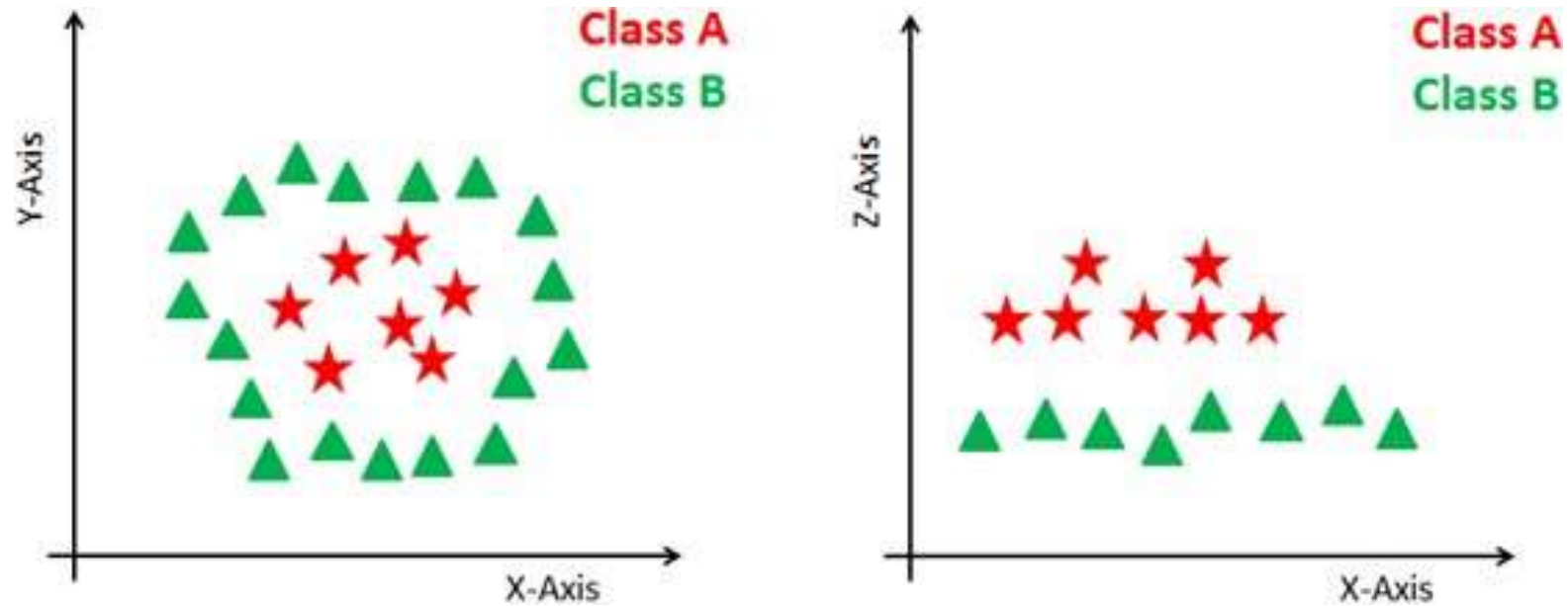
# How SVM works ?



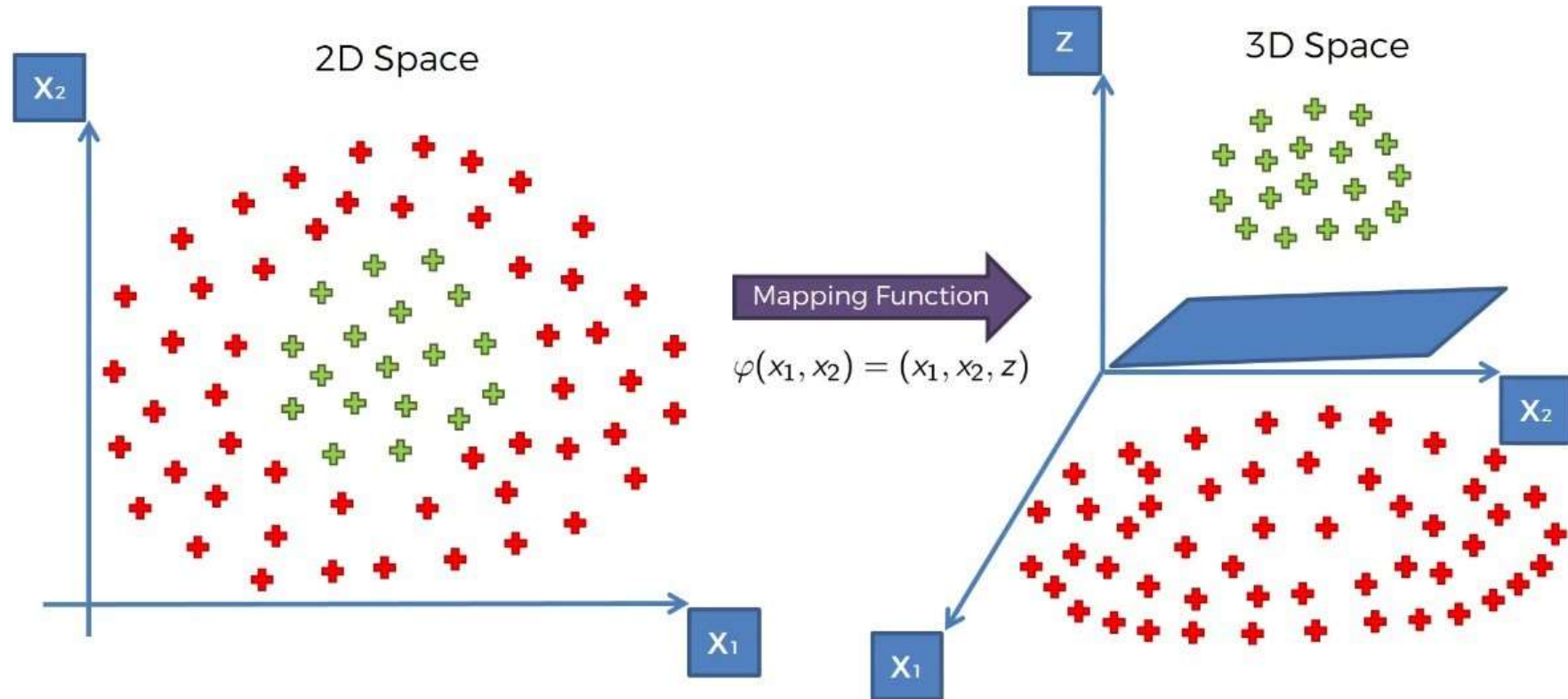
# Non-linear and inseparable planes

- Some problems can't be solved using linear hyperplane.
- In such situation, SVM uses a kernel trick to transform the input space to a higher dimensional space.
- The data points are plotted on the x-axis and z- axis (Z is the squared sum of both x and y:  $z=x^2+y^2$ ).
- Now you can easily segregate these points using linear separation.

# Non-linear and inseparable planes



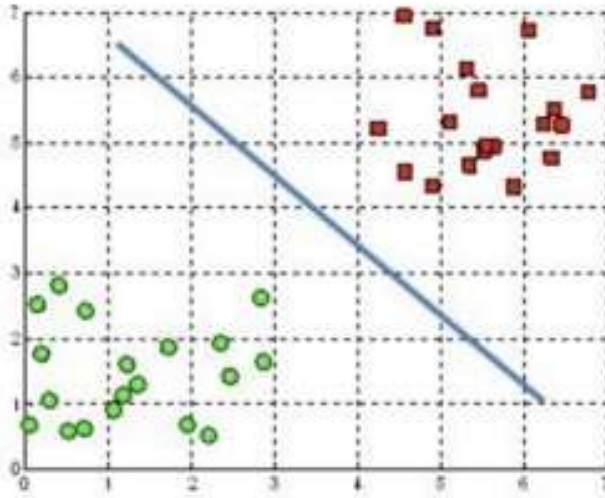
# High Dimensional Space Mapping



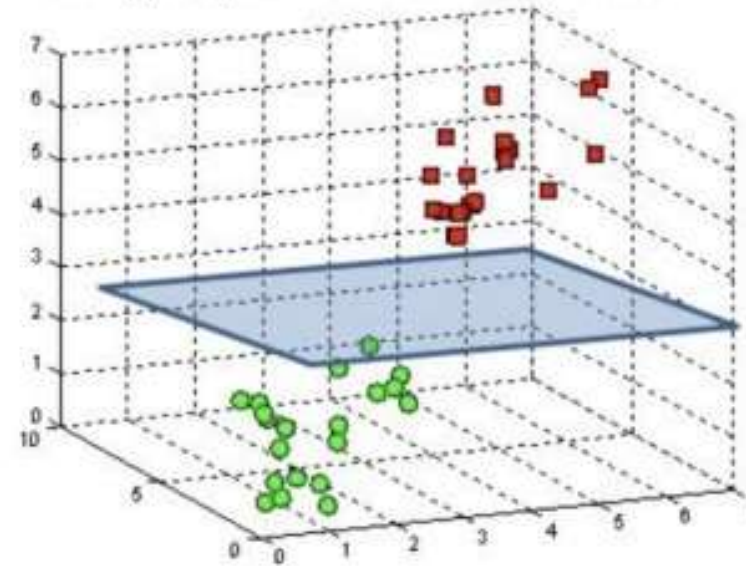


# High Dimensional Space Mapping

A hyperplane in  $\mathbb{R}^2$  is a line



A hyperplane in  $\mathbb{R}^3$  is a plane



# SVM Kernels

- The SVM algorithm is implemented in practice using a kernel. A kernel transforms an input data space into the required form.
- SVM uses a technique called the kernel trick. Here, the kernel takes a low-dimensional input space and transforms it into a higher dimensional space.
- In other words, you can say that it converts non-separable problem to separable problems by adding more dimension to it.
- It is most useful in non-linear separation problem. Kernel trick helps you to build a more accurate classifier.

# Kernel Types

- Linear Kernel
- Polynomial Kernel
- Radial Basis Function Kernel
- Sigmoid Kernel

Some of the common kernel functions for transforming from a lower dimension ' $i$ ' to a higher dimension ' $j$ ' used by different SVM implementations are as follows:

- Linear kernel: It is in the form  $K(\vec{x}_i, \vec{x}_j) = \vec{x}_i \cdot \vec{x}_j$
- Polynomial kernel: It is in the form  $K(\vec{x}_i, \vec{x}_j) = (\vec{x}_i \cdot \vec{x}_j + 1)^d$
- Sigmoid kernel: It is in the form  $K(\vec{x}_i, \vec{x}_j) = \tanh(k\vec{x}_i \cdot \vec{x}_j - \theta)$
- Gaussian RBF kernel: It is in the form  $K(\vec{x}_i, \vec{x}_j) = e^{\frac{-\|\vec{x}_i - \vec{x}_j\|^2}{2\sigma^2}}$

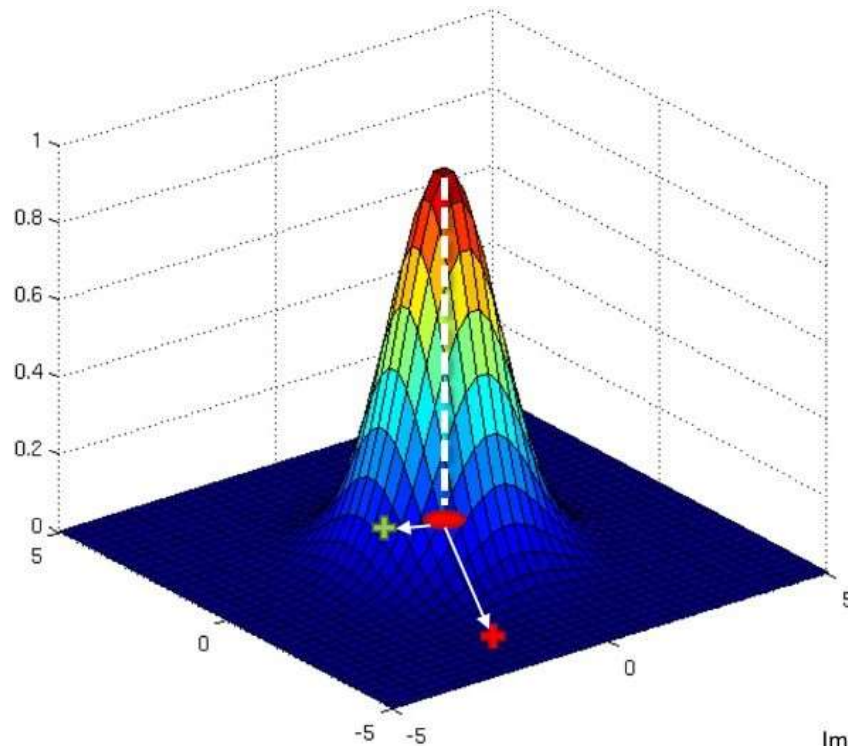
# Radial Basis Function Kernel

- The Radial basis function kernel is a popular kernel function commonly used in support vector machine classification. RBF can map an input space in infinite dimensional space.
- Here gamma is a parameter, which ranges from 0 to 1. A higher value of gamma will perfectly fit the training dataset, which causes over-fitting. Gamma=0.1 is considered to be a good default value.
- The value of gamma needs to be manually specified in the learning algorithm.

# Radial Basis Function Kernel

$$K(\vec{x}, \vec{l}^i) = e^{-\frac{\|\vec{x} - \vec{l}^i\|^2}{2\sigma^2}}$$

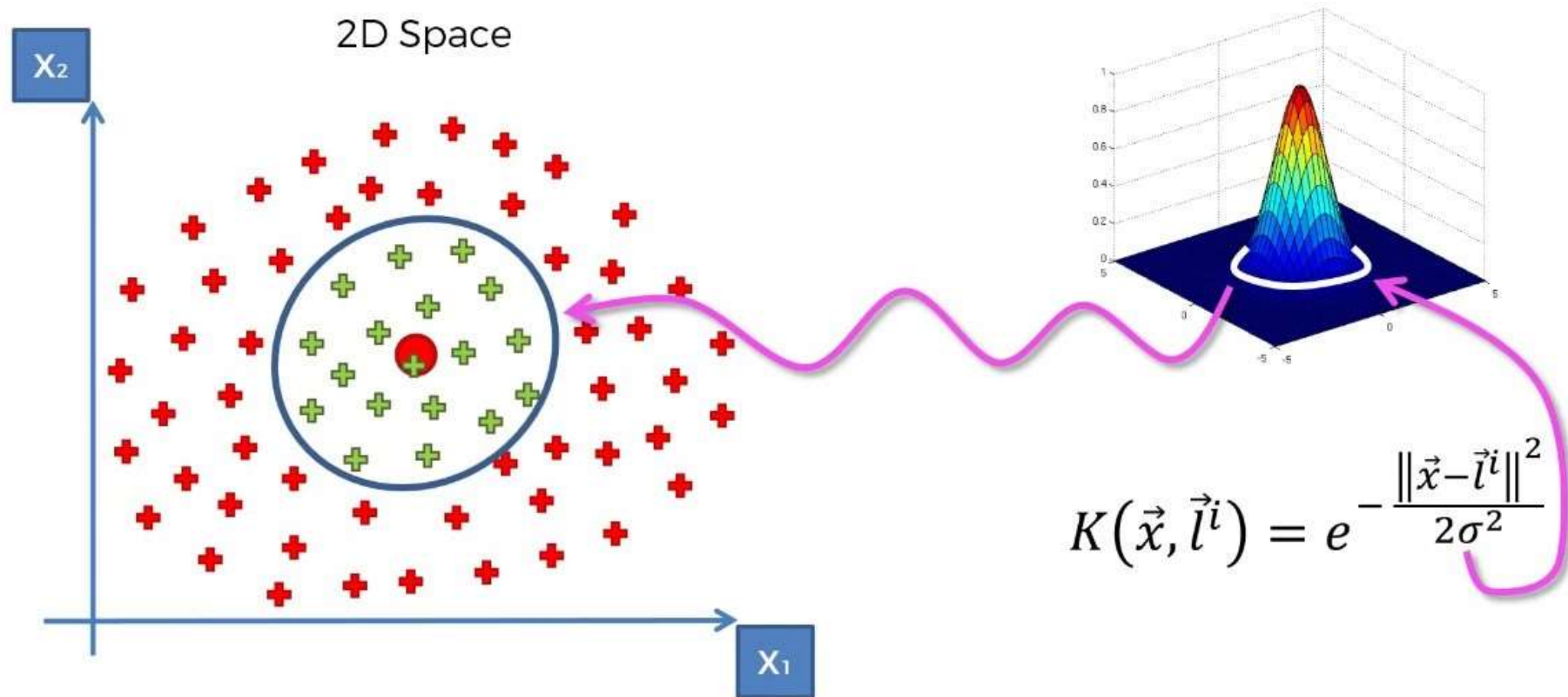
# Radial Basis Function Kernel



$$K(\vec{x}, \vec{l}^i) = e^{-\frac{\|\vec{x} - \vec{l}^i\|^2}{2\sigma^2}}$$

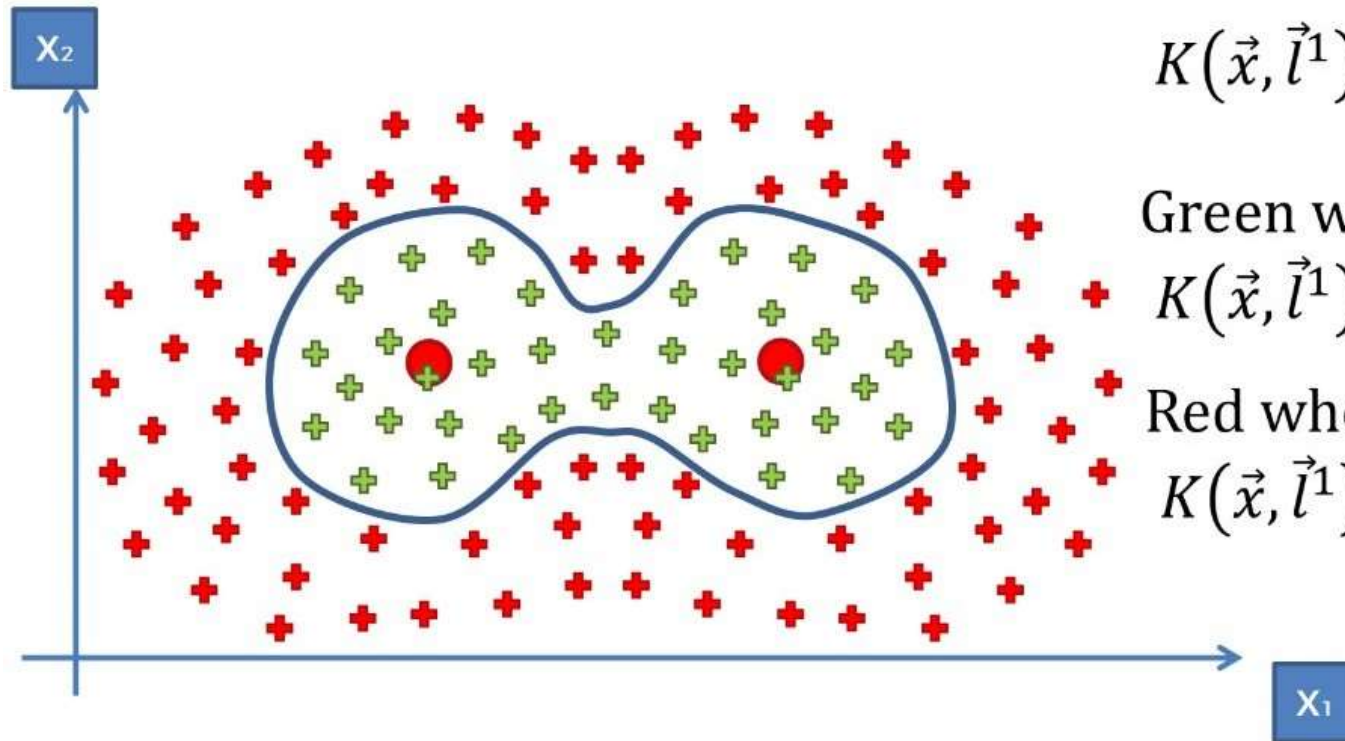
Image source: <http://www.cs.toronto.edu/~duvenaud/cookbook/index.html>

# Radial Basis Function Kernel





# Radial Basis Function Kernel



$$K(\vec{x}, \vec{l}^1) + K(\vec{x}, \vec{l}^2)$$

(Simplified Formula)

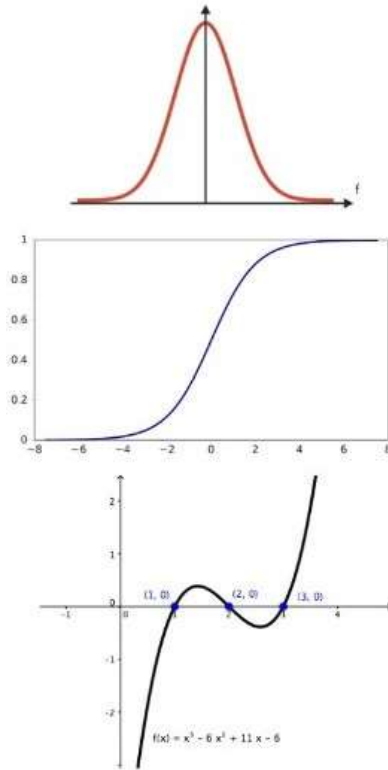
Green when:

$$K(\vec{x}, \vec{l}^1) + K(\vec{x}, \vec{l}^2) > 0$$

Red when:

$$K(\vec{x}, \vec{l}^1) + K(\vec{x}, \vec{l}^2) = 0$$

# Types of Kernels



Gaussian RBF Kernel

$$K(\vec{x}, \vec{l}^i) = e^{-\frac{\|\vec{x} - \vec{l}^i\|^2}{2\sigma^2}}$$

Sigmoid Kernel

$$K(X, Y) = \tanh(\gamma \cdot X^T Y + r)$$

Polynomial Kernel

$$K(X, Y) = (\gamma \cdot X^T Y + r)^d, \gamma > 0$$

### ***Strengths of SVM***

- SVM can be used for both classification and regression.
- It is robust, i.e. not much impacted by data with noise or outliers.
- The prediction results using this model are very promising.

### ***Weaknesses of SVM***

- SVM is applicable only for binary classification, i.e. when there are only two classes in the problem domain.
- The SVM model is very complex – almost like a black box when it deals with a high-dimensional data set. Hence, it is very difficult and close to impossible to understand the model in such cases.
- It is slow for a large dataset, i.e. a data set with either a large number of features or a large number of instances.
- It is quite memory-intensive.

# Rule-Based Classification

# Using IF-THEN Rules for Classification

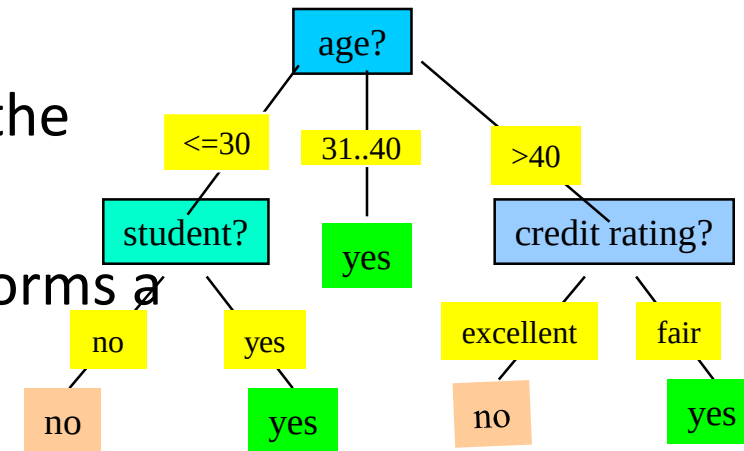
- Represent the knowledge in the form of IF-THEN rules

R: IF *age* = youth AND *student* = yes THEN *buys\_computer* = yes

- Rule antecedent/precondition vs. rule consequent
- Assessment of a rule: *coverage* and *accuracy*
  - $n_{\text{covers}}$  = # of tuples covered by R
  - $n_{\text{correct}}$  = # of tuples correctly classified by R
$$\text{coverage}(R) = n_{\text{covers}} / |D| \quad /* D: \text{training data set} */$$
$$\text{accuracy}(R) = n_{\text{correct}} / n_{\text{covers}}$$
- If more than one rule are triggered, need **conflict resolution**
  - Size ordering: assign the highest priority to the triggering rules that has the “toughest” requirement (i.e., with the *most attribute tests*)
  - Class-based ordering: decreasing order of *prevalence or misclassification cost per class*
  - Rule-based ordering (**decision list**): rules are organized into one long priority list, according to some measure of rule quality or by experts

# Rule Extraction from a Decision Tree

- Rules are *easier to understand* than large trees
- One rule is created *for each path* from the root to a leaf
- Each attribute-value pair along a path forms a conjunction: the leaf holds the class prediction
- Rules are mutually exclusive and exhaustive
- Example: Rule extraction from our *buys\_computer* decision-tree



IF *age* = young AND *student* = *no*

THEN *buys\_computer* = *no*

IF *age* = young AND *student* = *yes*

THEN *buys\_computer* = *yes*

IF *age* = mid-age

THEN *buys\_computer* = *yes*

IF *age* = old AND *credit\_rating* = *excellent*

THEN *buys\_computer* = *no*

IF *age* = old AND *credit\_rating* = *fair*

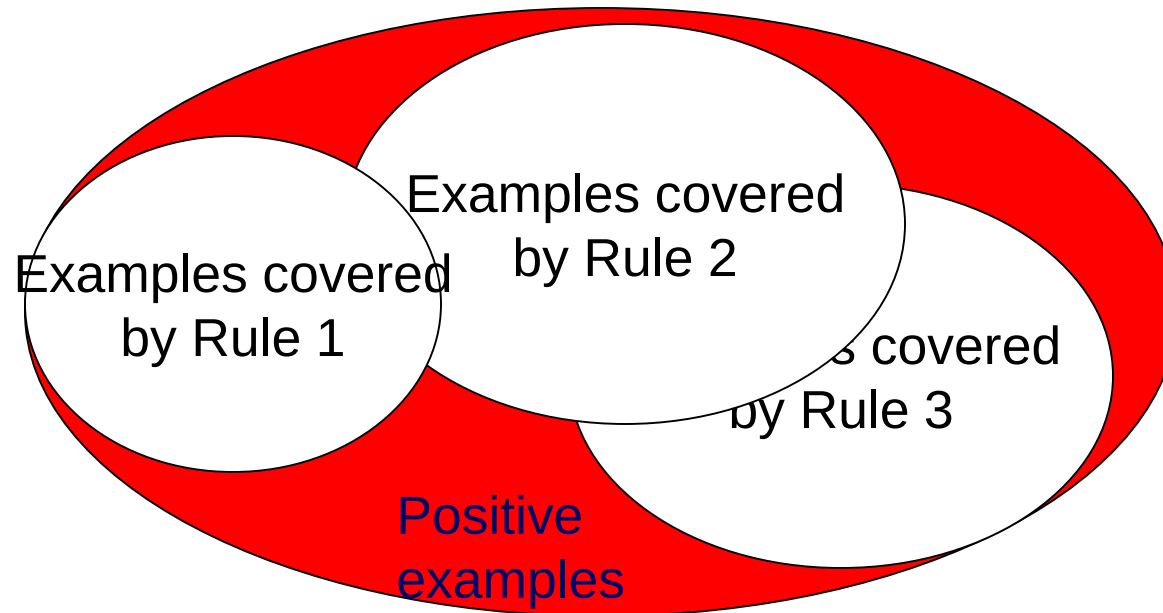
THEN *buys\_computer* = *yes*

# Rule Induction: Sequential Covering Method

- Sequential covering algorithm: Extracts rules directly from training data
- Typical sequential covering algorithms: FOIL, AQ, CN2, RIPPER
- Rules are learned *sequentially*, each for a given class  $C_i$  will cover many tuples of  $C_i$  but none (or few) of the tuples of other classes
- Steps:
  - Rules are learned one at a time
  - Each time a rule is learned, the tuples covered by the rules are removed
  - Repeat the process on the remaining tuples until *termination condition*, e.g., when no more training examples or when the quality of a rule returned is below a user-specified threshold
- Comp. w. decision-tree induction: learning a set of rules *simultaneously*

# Sequential Covering Algorithm

**while** (enough target tuples left)  
    generate a rule  
    remove positive target tuples satisfying this rule





# Rule Generation

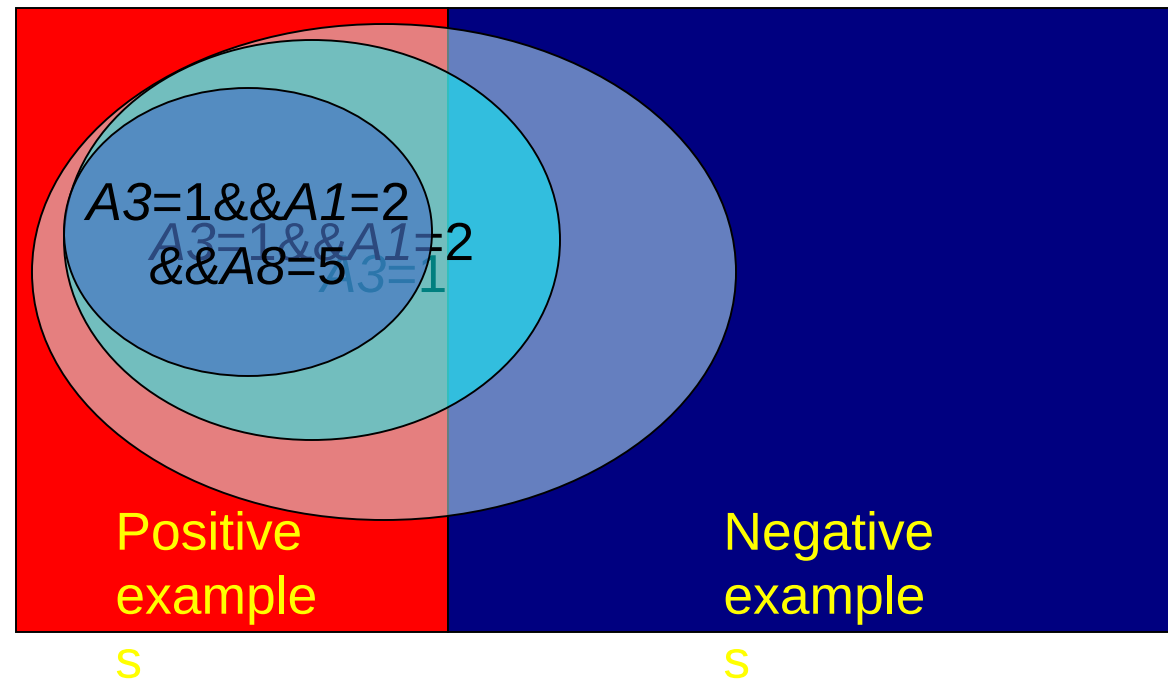
- To generate a rule

**while**(true)

find the best predicate  $p$

**if**  $\text{foil-gain}(p) > \text{threshold}$  **then** add  $p$  to current rule

**else break**



# How to Learn-One-Rule?

- Start with the *most general rule* possible: condition = empty
- *Adding new attributes* by adopting a greedy depth-first strategy
  - Picks the one that most improves the rule quality
- Rule-Quality measures: consider both coverage and accuracy
  - Foil-gain (in FOIL & RIPPER): assesses info\_gain by extending condition

$$FOIL\_Gain = pos' \times (\log_2 \frac{pos'}{pos' + neg'} - \log_2 \frac{pos}{pos + neg})$$

- favors rules that have high accuracy and cover many positive tuples
- Rule pruning based on an independent set of test tuples

$$FOIL\_Prune(R) = \frac{pos - neg}{pos + neg}$$

Pos/neg are # of positive/negative tuples covered by R.

If  $FOIL\_Prune$  is higher for the pruned version of R, prune R